Multi-region input-output (MRIO) tables

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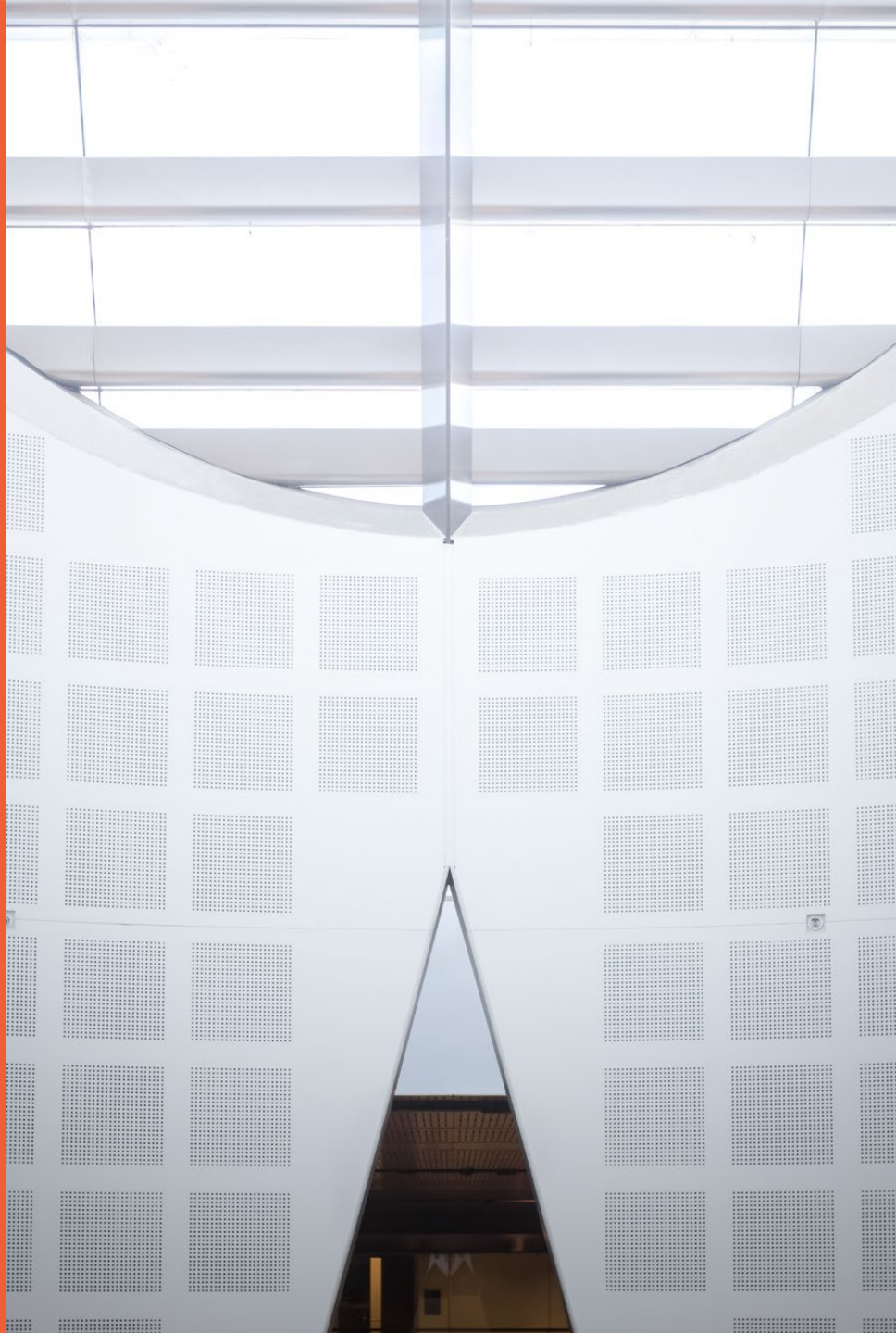


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Recap – Week 6

- The **Q** block forms the basis for a production-based account of the indicator in question

- The **m** matrix forms the basis for a consumption-based account of the indicator in questions

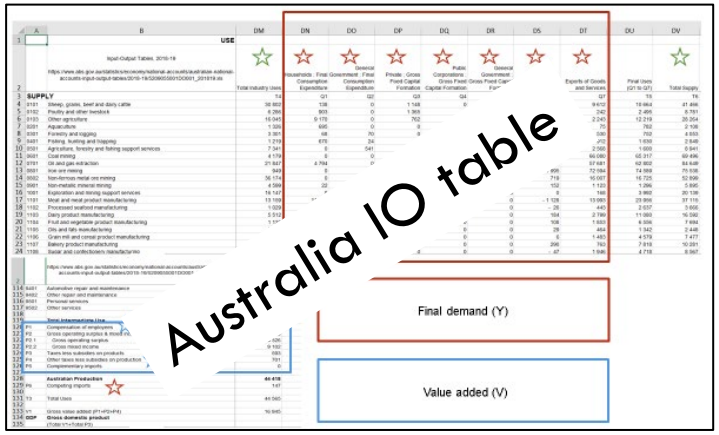
- Production-based vs consumption-based accounting for an environmental or social indicator

$$\boldsymbol{Footprint = \mathbf{m}y = \mathbf{q}\mathbf{L}y}$$

Types of input-output tables

Single-region input-output tables

Tables that feature information for one region only.



		Afghanistan AFG Industries Agriculture	Afghanistan AFG Industries Fishing	Afghanistan AFG Industries Mining and	Afghanistan AFG Industries Food & Be	Afghanistan AFG Industries Textiles an	Afghanistan AFG Industries Wood and	Afghanistan AFG Industries Petroleum,	Afghanistan AFG Industries Metal Prod	Afghanistan AFG Industries Electrical a	Afghanistan AFG Industries Transport	Afghanistan AFG Industries Other Man	A AFG Ir R
Environmental	Water use (m3)	1.84E+10	3.21E+07	1.48E+04	1.83E+09	1.85E+04	4.06E+04	1.21E+05	4.78E+04	1.20E+05	6.43E+04	4.04E+08	
	Nitrogen emissions (Gg)	3.21E+02	8.31E-04	1.31E+00	3.67E-02	5.22E-03	1.41E-02	7.58E-02	1.58E-02	3.96E-02	1.99E-02	5.70E-03	
	Energy use (TJ)	2.64E+01	7.69E-01	1.15E+01	5.91E+01	1.08E+01	2.78E+01	7.76E+01	3.15E+01	8.01E+01	4.34E+01	1.23E+01	
	CO2-e (Gg)	1.11E+04	2.80E+01	1.40E+03	5.91E+01	3.18E+01	4.15E+01	1.34E+02	7.96E+01	7.09E+01	5.02E+01	3.27E+01	
Social	Male Employment (FTE)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	
	Female Employment (FTE)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	
Economic	Value Added (USD)	4.36E+05	1.99E+04	3.13E+04	3.29E+05	7.87E+04	2.52E+05	5.52E+05	2.39E+05	6.46E+05	1.91E+05	9.47E+04	

Multi-region input-output tables

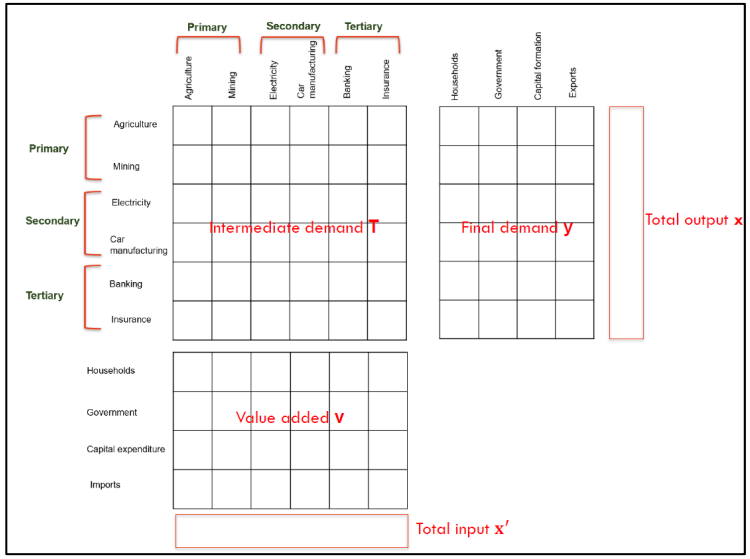
Tables that feature information for more than one region.

Single region input-output tables (SRIO)

Regions: ONE ONLY
Sectors: Can vary

Aggregated SRIO tables (low sectoral resolution), for example the input-output table for Norfolk Island only has a total of 18 sectors, with Agricultural activity captured in one 'Agriculture' sector.

Disaggregated SRIO tables (high sectoral resolution), for example the input-output table for Australia has sectors with specific detail on agriculture including 'Sheep, grains, beef and dairy cattle', 'Poultry and other livestock', 'Other agriculture'.

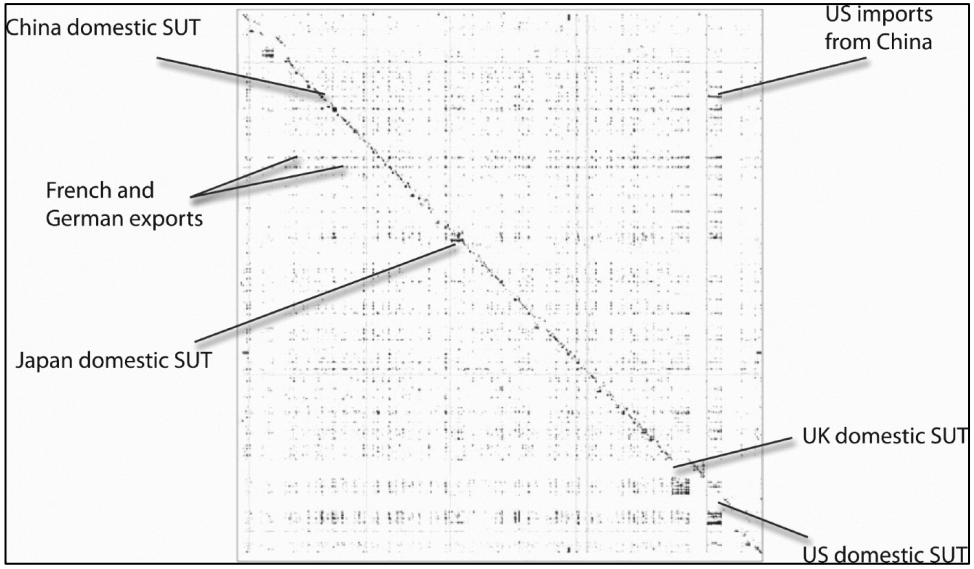


Multi-region input-output tables (MRIO)

Regions: More than one
Sectors: Can vary

Sub-national MRIO table: Contains data on different regions within a country, for example states & territories, local government areas.

Global MRIO table: Contains data on different countries, for example UK, China, Japan, France, Germany, US.



Lenzen, M, Moran, D, Kanemoto, K, & Geschke, A, 2013, 'BUILDING EORA: A GLOBAL MULTI-REGION INPUT-OUTPUT DATABASE AT HIGH COUNTRY AND SECTOR RESOLUTION', *Economic Systems Research*, vol. 25, no. 1, pp. 20–49, doi: 10.1080/09535314.2013.769938.

Multi-region input-output tables (MRIO)

USD$		Australia						New Zealand					Fiji			TOTAL Intermediate output	AUS		NZL		FJI		ROW	TOTAL Final Demand	TOTAL OUTPUT
		Ag	Mining	Constr.	Manuf.	Retail	Banking	Sheep	Other Ag.	Manuf.	Tourism	Oth. Services	Fishing	Tourism	Other		HH	GOV	HH	GOV	HH	GOV	Exports		
Australia	Agriculture																								
	Mining																								
	Construction																								
	Manufacturing																								
	Retail																								
	Banking																								
NZ	Sheep																								
	Other Agriculture																								
	Manufacturing																								
	Tourism																								
	Other services																								
Fiji	Fishing																								
	Tourism																								
	Other																								
TOTAL Intermediate input																									
AUS	HH																								
	GOV																								
NZL	HH																								
	GOV																								
FJI	HH																								
	GOV																								
ROW	Imports																								
TOTAL Value-added																									

$T_{AUS-AUS}$ $T_{AUS-NZL}$ $T_{AUS-FJI}$ $Y_{AUS-AUS}$ $Y_{AUS-NZL}$ $Y_{AUS-FJI}$

$T_{NZL-AUS}$ $T_{NZL-NZL}$ $T_{NZL-FJI}$ $Y_{NZL-AUS}$ $Y_{NZL-NZL}$ $Y_{NZL-FJI}$

$T_{FJI-AUS}$ $T_{FJI-NZL}$ $T_{FJI-FJI}$ $Y_{FJI-AUS}$ $Y_{FJI-NZL}$ $Y_{FJI-FJI}$

$V_{AUS-AUS}$ $V_{AUS-NZL}$ $V_{AUS-FJI}$

$V_{NZL-AUS}$ $V_{NZL-NZL}$ $V_{NZL-FJI}$

$V_{FJI-AUS}$ $V_{FJI-NZL}$ $V_{FJI-FJI}$



Output

(sheep exported from NZ to AUS)



Input

(expenditure on imports from AUS Agriculture to NZ sheep)

Multi-region input-output tables (MRIO)

USD$		Australia						New Zealand					Fiji			TOTAL Intermediate output	AUS		NZL		FJI		ROW	TOTAL Final Demand	TOTAL OUTPUT
		Ag	Mining	Constr.	Manuf.	Retail	Banking	Sheep	Other Ag.	Manuf.	Tourism	Oth. Services	Fishing	Tourism	Other		HH	GOV	HH	GOV	HH	GOV	Exports		
Australia	Agriculture	AUS National IO Table						Exports from AUS to NZL $T_{AUS-NZL}$					Exports from AUS to FJI $T_{AUS-FJI}$				$Y_{AUS-AUS}$		$Y_{AUS-NZL}$		$Y_{AUS-FJI}$				
	Mining																								
	Construction																								
	Manufacturing																								
	Retail																								
	Banking																								
NZ	Sheep	Exports from NZL to AUS $T_{NZL-AUS}$						NZ National IO Table					Exports from NZ to FJI $T_{NZL-FJI}$				$Y_{NZL-AUS}$		$Y_{NZL-NZL}$		$Y_{NZL-FJI}$				
	Other Agriculture																								
	Manufacturing																								
	Tourism																								
	Other services																								
Fiji	Fishing	Exports from FJI to AUS $T_{FJI-AUS}$						Exports from FJI to NZL $T_{FJI-NZL}$					FJI National IO Table				$Y_{FJI-AUS}$		$Y_{FJI-NZL}$		$Y_{FJI-FJI}$				
	Tourism																								
	Other																								
TOTAL Intermediate input																									
AUS	HH	$V_{AUS-AUS}$						$V_{AUS-NZL}$					$V_{AUS-FJI}$												
	GOV																								
NZL	HH	$V_{NZL-AUS}$						$V_{NZL-NZL}$					$V_{NZL-FJI}$												
	GOV																								
FJI	HH	$V_{FJI-AUS}$						$V_{FJI-NZL}$					$V_{FJI-FJI}$												
	GOV																								
ROW	Imports																								
TOTAL Value-added																									

Diagonal matrices:
National input-output tables

Off-diagonal matrices:
Trade matrices

Australia → China → Vietnam → New Zealand

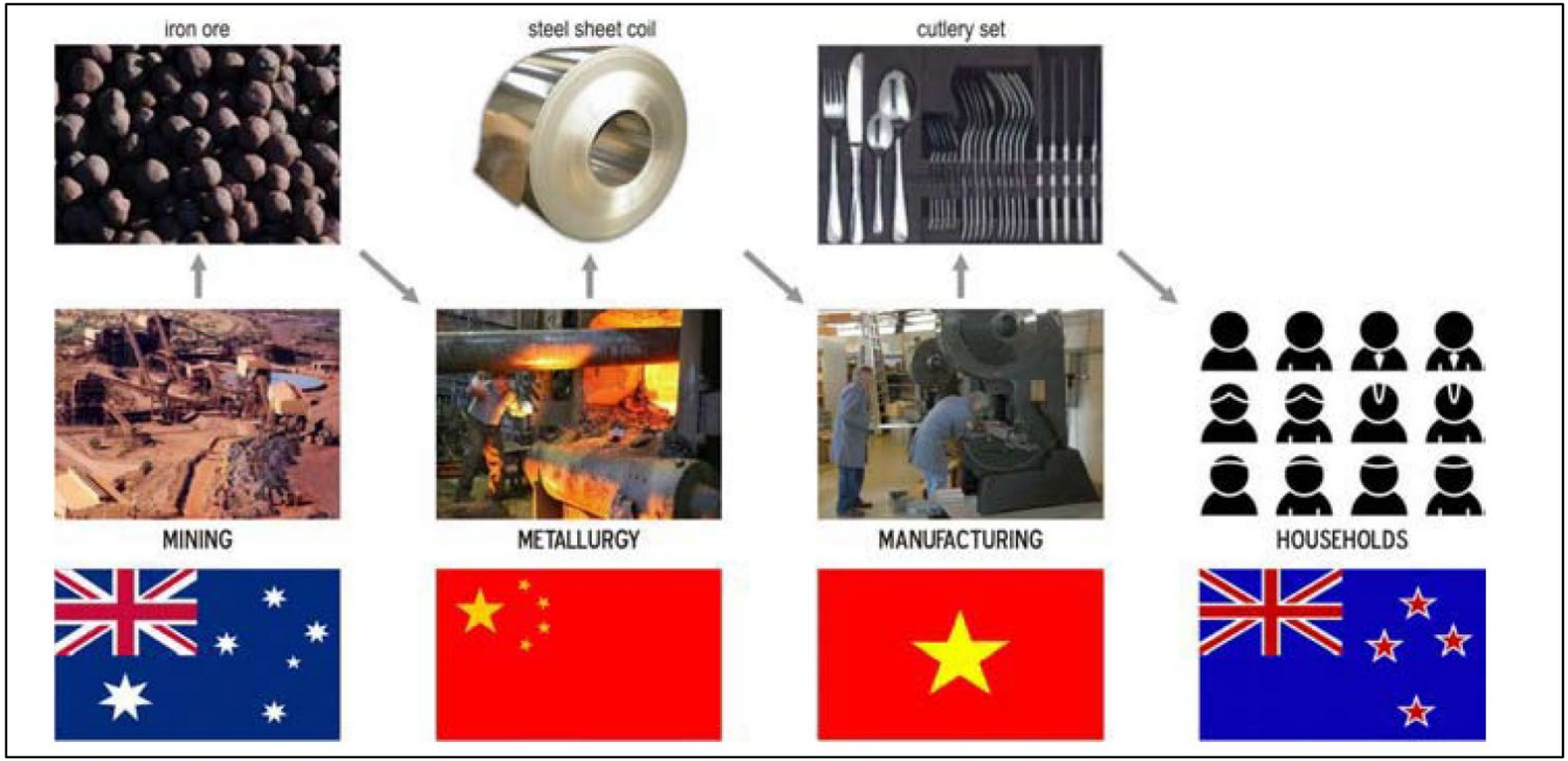
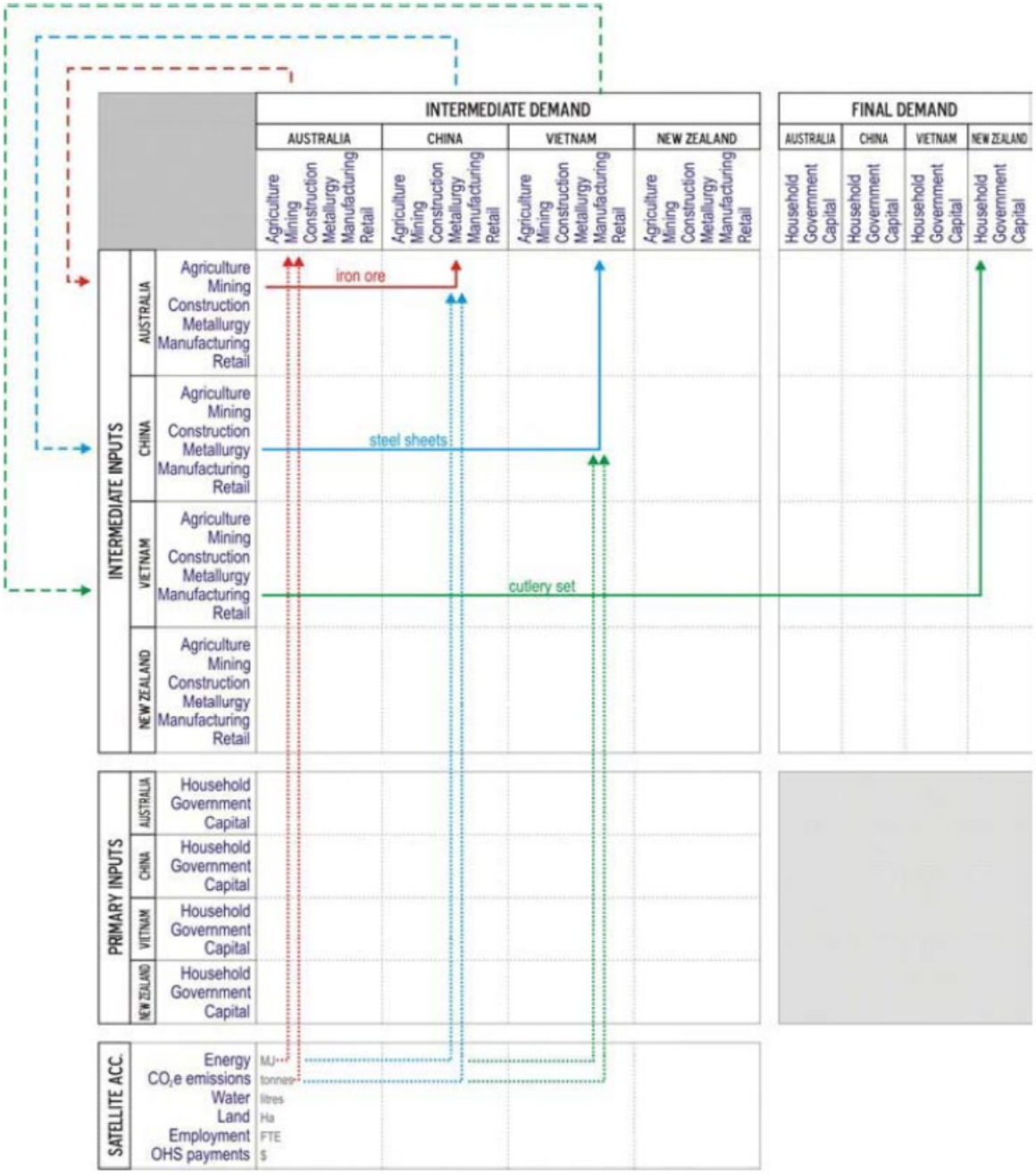
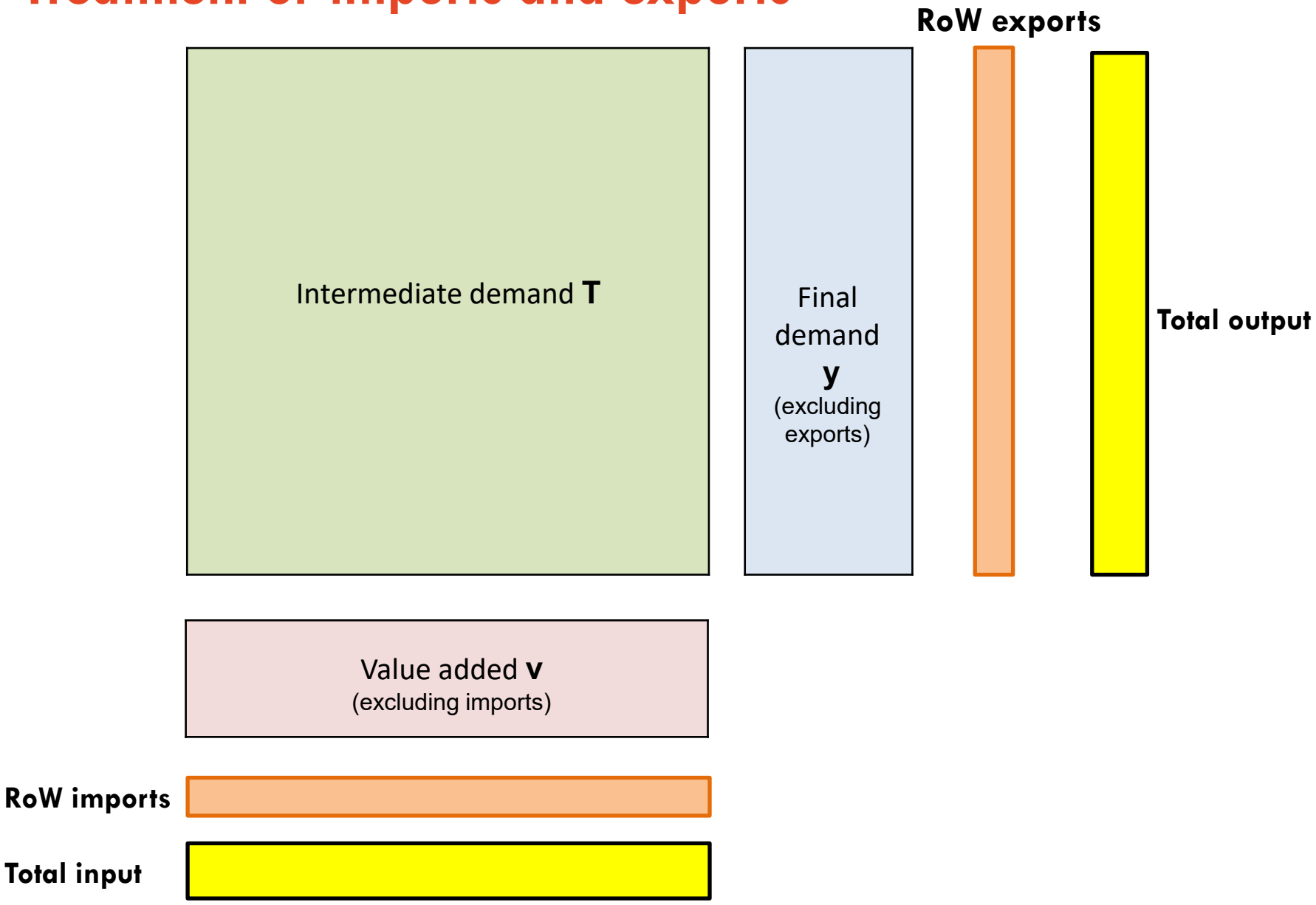


Figure 8.2: Illustrative supply chain which includes the purchase of a cutlery set by a household consumer in New Zealand, which was manufactured in Vietnam using steel sourced from China, which was produced with iron ore mined in Australia.


INTERMEDIATE DEMAND
AUSTRALIA
CHINA
VIETNAM
NEW ZEALAND
Agriculture
Mining
Construction
Metallurgy
Manufacturing
Retail
FINAL DEMAND
Household
Government
Capital
INTERMEDIATE INPUTS
PRIMARY INPUTS
iron ore
steel sheets
cutlery set
SATELLITE ACC.
Energy
CO2e emissions
Water
Land
Employment
OHS payments
MJ
tonnes
litres
Ha
FTE
$

Treatment of imports and exports



Treatment of imports and exports

USD$		Australia						New Zealand					Fiji			TOTAL Intermediate output	AUS		NZL		FJI		ROW	TOTAL Final Demand	TOTAL OUTPUT
		Ag	Mining	Constr.	Manuf.	Retail	Banking	Sheep	Other Ag.	Manuf.	Tourism	Oth. Services	Fishing	Tourism	Other		HH	GOV	HH	GOV	HH	GOV	Exports		
Australia	Agriculture	$T_{AUS-AUS}$						$T_{AUS-NZL}$					$T_{AUS-FJI}$				$Y_{AUS-AUS}$		$Y_{AUS-NZL}$		$Y_{AUS-FJI}$				
	Mining																								
	Construction																								
	Manufacturing																								
	Retail																								
	Banking																								
NZ	Sheep	$T_{NZL-AUS}$						$T_{NZL-NZL}$					$T_{NZL-FJI}$				$Y_{NZL-AUS}$		$Y_{NZL-NZL}$		$Y_{NZL-FJI}$				
	Other Agriculture																								
	Manufacturing																								
	Tourism																								
	Other services																								
Fiji	Fishing	$T_{FJI-AUS}$						$T_{FJI-NZL}$					$T_{FJI-FJI}$				$Y_{FJI-AUS}$		$Y_{FJI-NZL}$		$Y_{FJI-FJI}$				
	Tourism																								
	Other																								
TOTAL Intermediate input																									
AUS	HH	$V_{AUS-AUS}$						$V_{AUS-NZL}$					$V_{AUS-FJI}$												
	GOV																								
NZL	HH	$V_{NZL-AUS}$						$V_{NZL-NZL}$					$V_{NZL-FJI}$												
	GOV																								
FJI	HH	$V_{FJI-AUS}$						$V_{FJI-NZL}$					$V_{FJI-FJI}$												
	GOV																								
ROW	Imports																								
TOTAL Value-added																									

Exports to countries other than Australia, New Zealand, and Fiji

Imports from countries other than Australia, New Zealand, and Fiji

Calculating imported and exported footprints

(extension content)

$$\boldsymbol{Footprint = my = qLy}$$

The **Q** block provides us with the production-based account for each indicator for each sector in each country. Let's separate the direct intensities matrix **q** into 2 matrices:

$\mathbf{q}_{aus}$

$\mathbf{q}_{oth}$

The MRIO provides us with final demand **y** for each sector in each country. Let's separate the **y** matrix into 2 matrices:

$\mathbf{y}_{aus}$

$\mathbf{y}_{oth}$

domestic footprint	$\mathbf{f}_{dom} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{aus}$
exported footprint	$\mathbf{f}_{exp} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{oth}$
imported footprint	$\mathbf{f}_{imp} = \mathbf{q}_{oth} * \mathbf{L} * \mathbf{y}_{aus}$

Calculating imported and exported footprints

(extension content)

domestic footprint	$\mathbf{f}_{dom} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{aus}$
exported footprint	$\mathbf{f}_{exp} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{oth}$
imported footprint	$\mathbf{f}_{imp} = \mathbf{q}_{oth} * \mathbf{L} * \mathbf{y}_{aus}$

$\mathbf{y}_{aus}$

AUS		NZL		FJI		ROW
HH	GOV	HH	GOV	HH	GOV	Exports
2	2344	0	0	0	0	0
1	2611	0	0	0	0	0
3789	4785	0	0	0	0	0
4756	3256	0	0	0	0	0
2564	1857	0	0	0	0	
957	654	0	0	0	0	
29	100	0	0	0	0	
50	60	0	0	0	0	
60	50	0	0	0	0	0
300	275	0	0	0	0	0
80	2	0	0	0	0	0
11	70	0	0	0	0	0
500	84	0	0	0	0	0
105	44	0	0	0	0	0

Note the zeros

$\mathbf{y}_{oth}$

AUS		NZL		FJI		ROW
HH	GOV	HH	GOV	HH	GOV	Exports
0	0	5	6	10	68	5648
0	0	23	325	1	514	7601
0	0	7	523	0	23	3561
0	0	74	500	0	45	3540
	0	66	30	20	8	1132
	0	45	50	45	14	1954
	0	9	50	12	100	62
	0	20	95	8	110	24
0	0	43	40	40	47	84
0	0	60	6	60	8	257
0	0	50	21	50	54	22
0	0	14	23	100	15	58
0	0	39	3	5	76	47
0	0	26	26	300	207	33

Calculating imported and exported footprints

(extension content)

domestic footprint	$\mathbf{f}_{dom} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{aus}$
exported footprint	$\mathbf{f}_{exp} = \mathbf{q}_{aus} * \mathbf{L} * \mathbf{y}_{oth}$
imported footprint	$\mathbf{f}_{imp} = \mathbf{q}_{oth} * \mathbf{L} * \mathbf{y}_{aus}$

Qaus

Australia						New Zealand					Fiji		
Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Sheep	Other Agriculture	Manufacturing	Tourism	Other services	Fishing	Tourism	Other
40	100	16	18	7	9	0	0	0	0	0	0	0	0

Note the zeros

Qoth

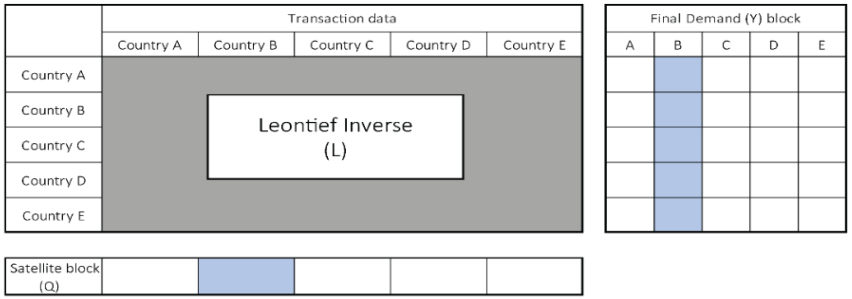
Australia						New Zealand					Fiji		
Agriculture	Mining	Construction	Manufacturing	Retail	Banking	Sheep	Other Agriculture	Manufacturing	Tourism	Other services	Fishing	Tourism	Other
0	0	0	0	0	0	10	5	3	6	2	3	9	13

Note the zeros

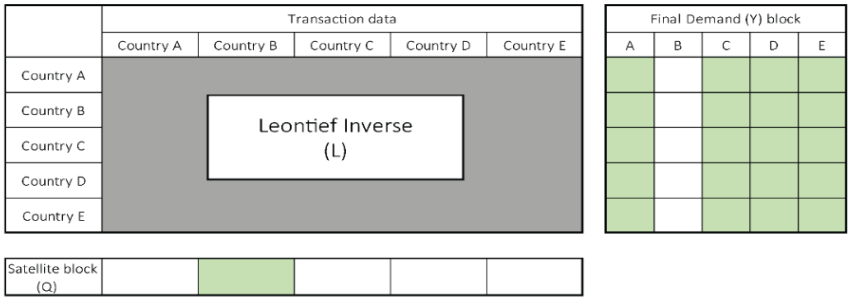
Calculating imported and exported footprints

(extension content)

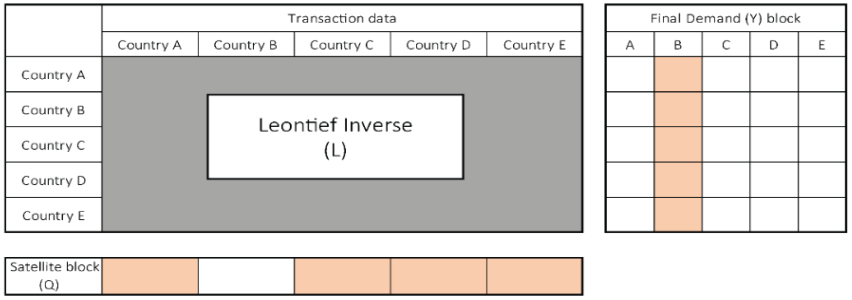
(a) Country B's domestic footprint is calculated using the Leontief inverse and the cells colour-coded blue



(b) Country B's exported footprint is calculated using the Leontief inverse and the cells colour-coded green



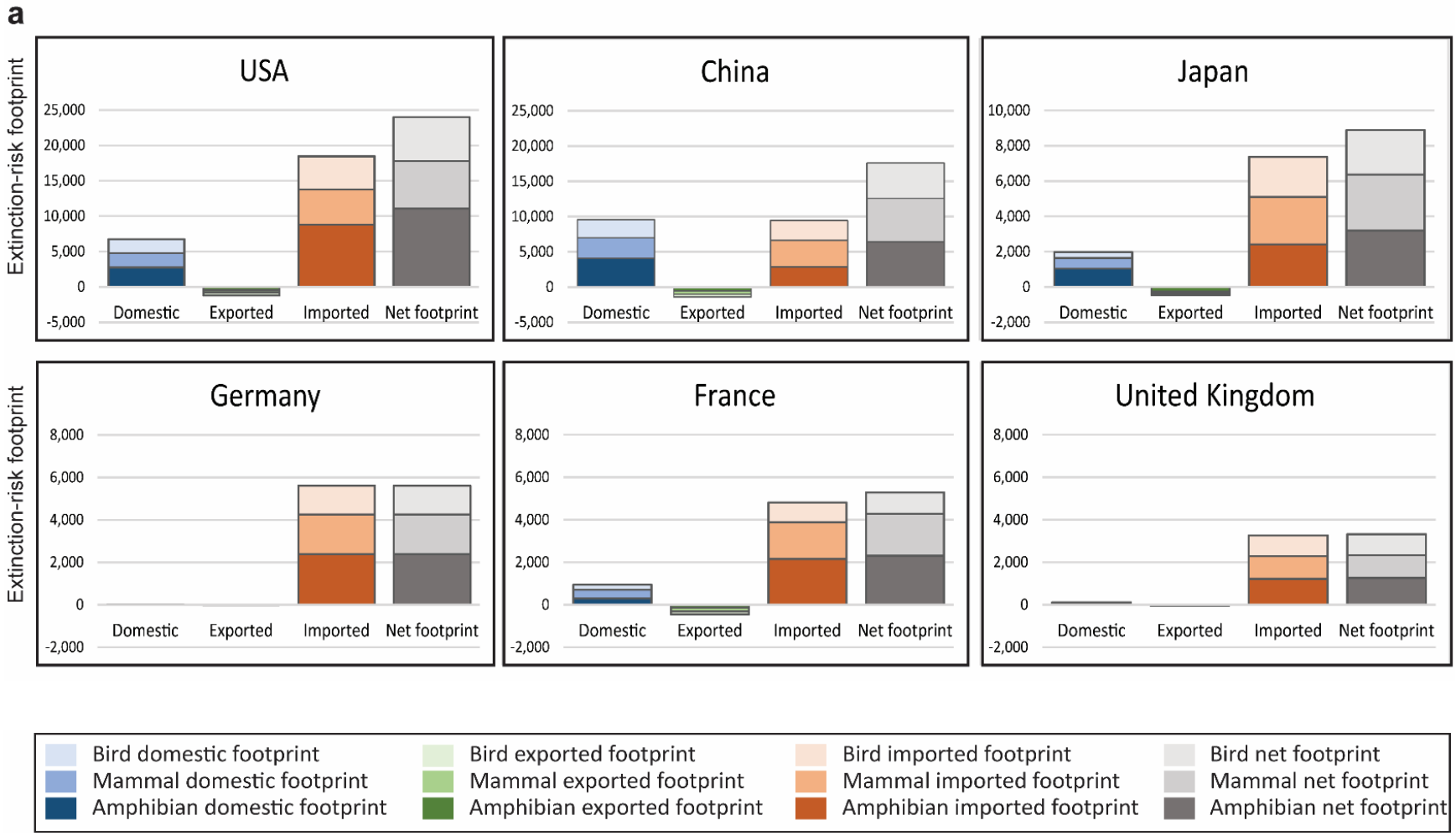
(c) Country B's imported footprint is calculated using the Leontief inverse and the cells colour-coded orange



A simplified illustration of the use of input-output analysis to calculate three sub-footprints for each country in a multi-region input-output table. The Leontief Inverse (**L**) matrix contains information on the economic interdependencies in this simple 5 country economy, which along with a 5x5 final demand (**Y**) matrix, and a 1x5 satellite block (**Q**), can be used to calculate 3 different footprints for each country: (a) the domestic footprint, (b) the exported footprint, and (c) the imported footprint.

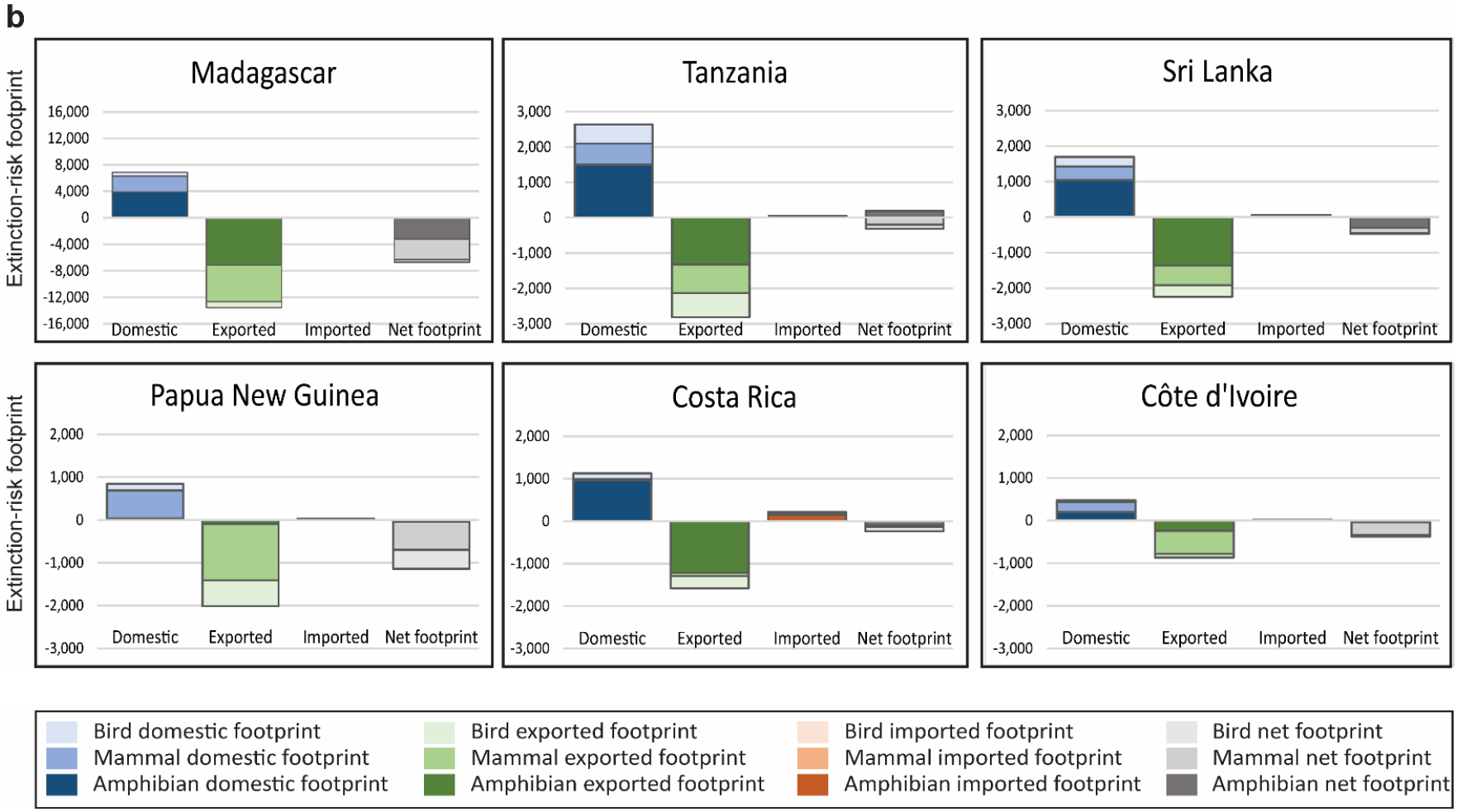
Irwin, A, Geschke, A, Brooks, TM, Siikamäki, J, Mair, L & Strassburg, BBN 2022, 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports*, vol. 12, no. 1, pp. 5861.

Extinction-risk footprint profile by country



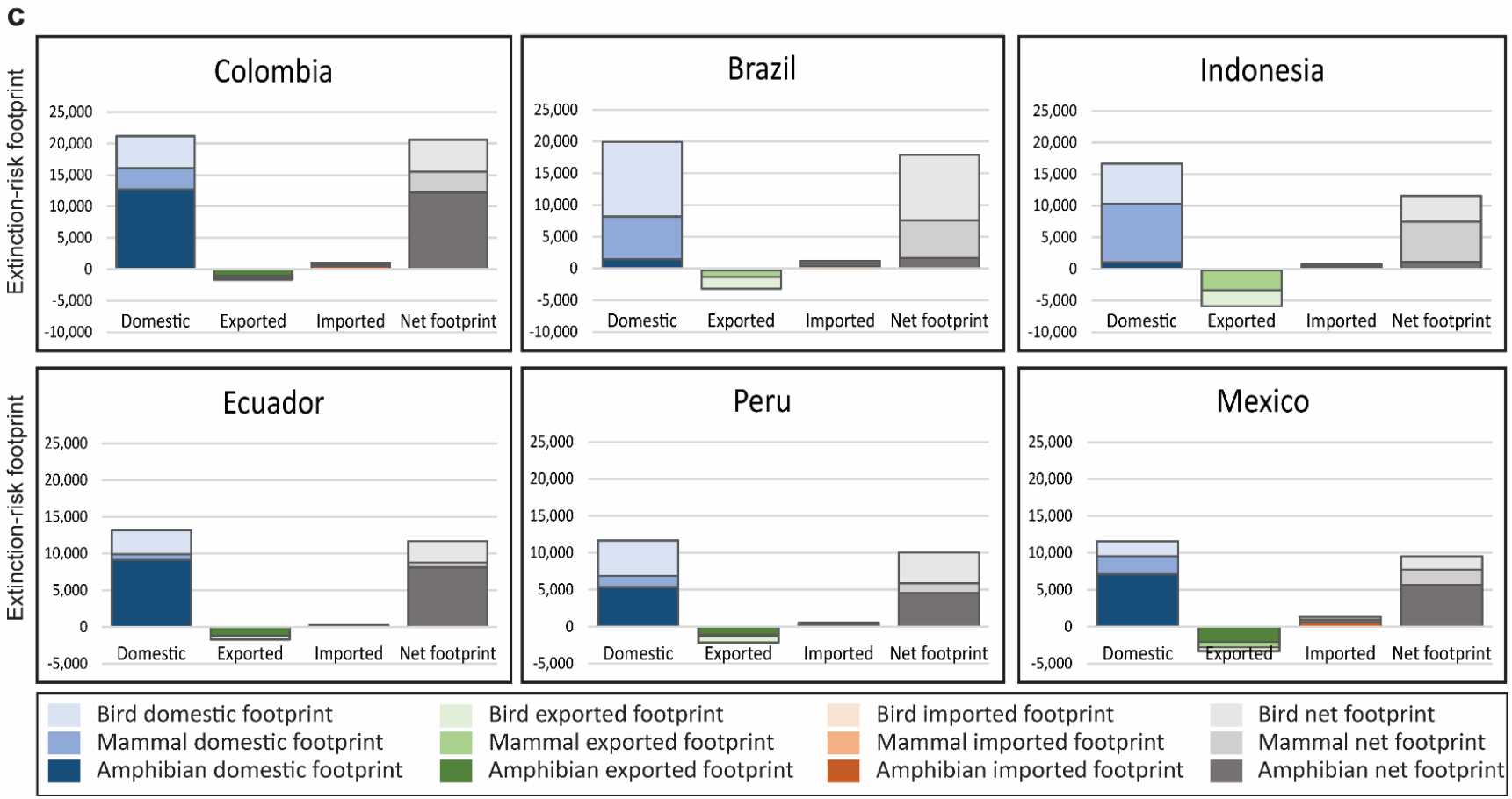
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Extinction-risk footprint profile by country



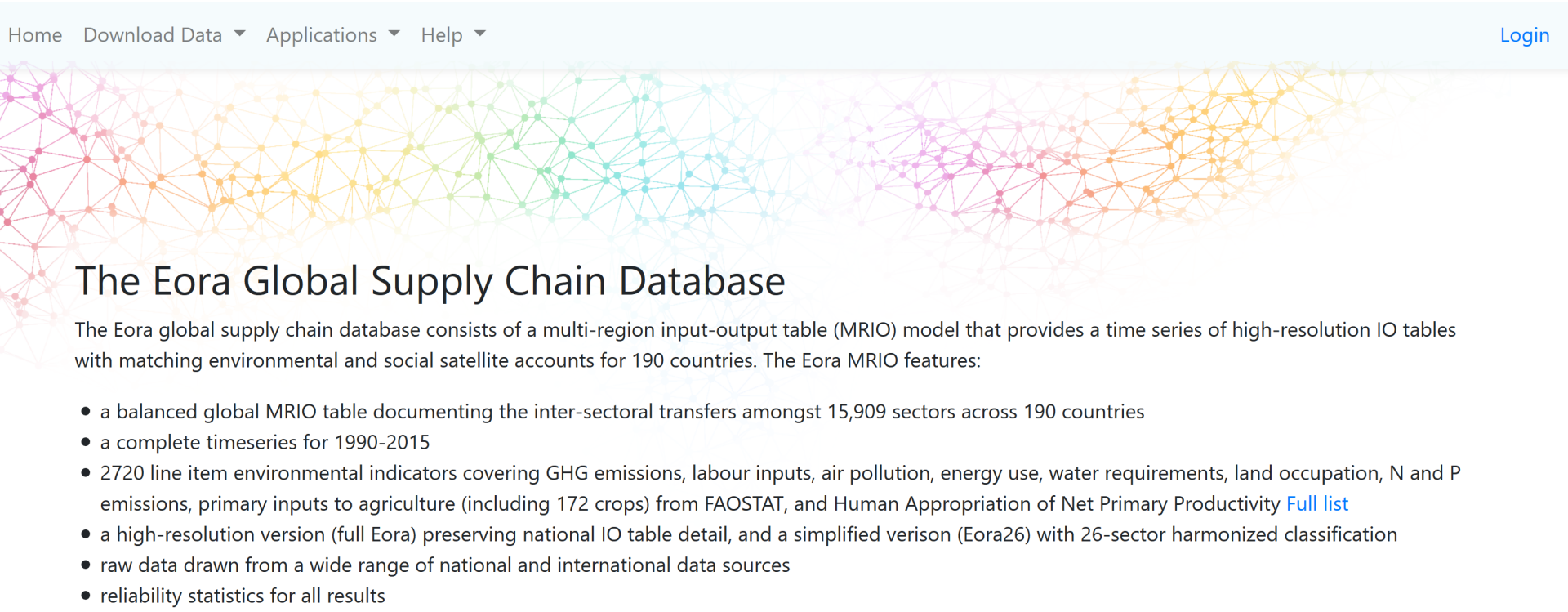
Irwin, A, Geschke, A, Brooks, TM, Siikamäki, J, Mair, L & Strassburg, BBN 2022, 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports*, vol. 12, no. 1, pp. 5861.

Extinction-risk footprint profile by country



Irwin, A, Geschke, A, Brooks, TM, Siikamäki, J, Mair, L & Strassburg, BBN 2022, 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports*, vol. 12, no. 1, pp. 5861.

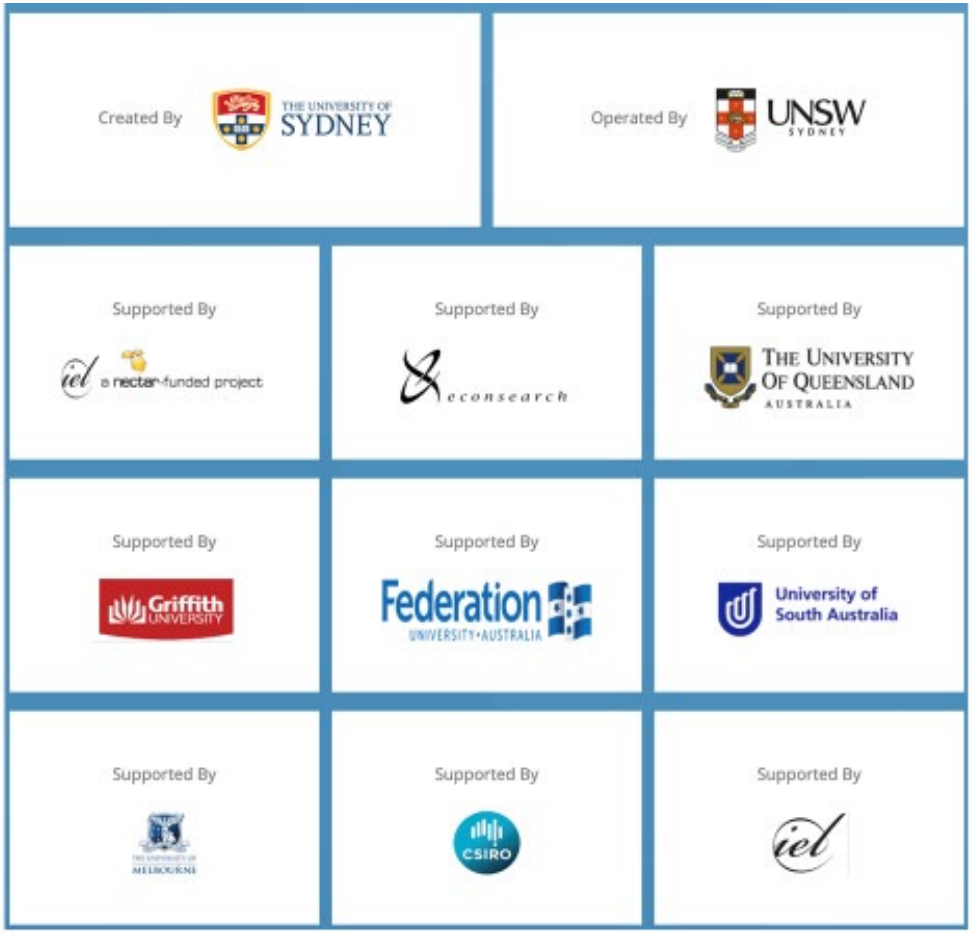
Global multi-regional input-output table



worldmrio.com

Constructing input-output tables for research using the IELab platform





https://ielab.info/

3 steps to creating an input-output table

1. Define your parameters: How many sectors? How many regions? How many years? Which sectors? Which regions? Which data sources do you want to pull the data from?

2. Compile the data based on these input parameters

3. Produce the final input-output table

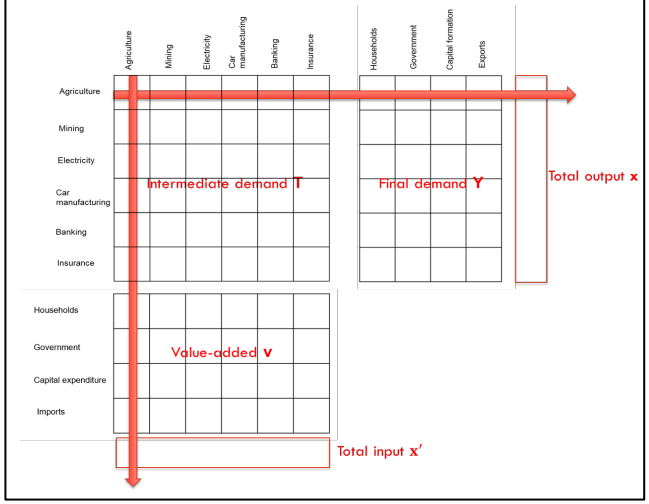
1. Define parameters:

- Regional/Sectoral Structure
- Data Sources (survey, non-survey, other IO-tables, …)

2. Compilation



3. Final Table



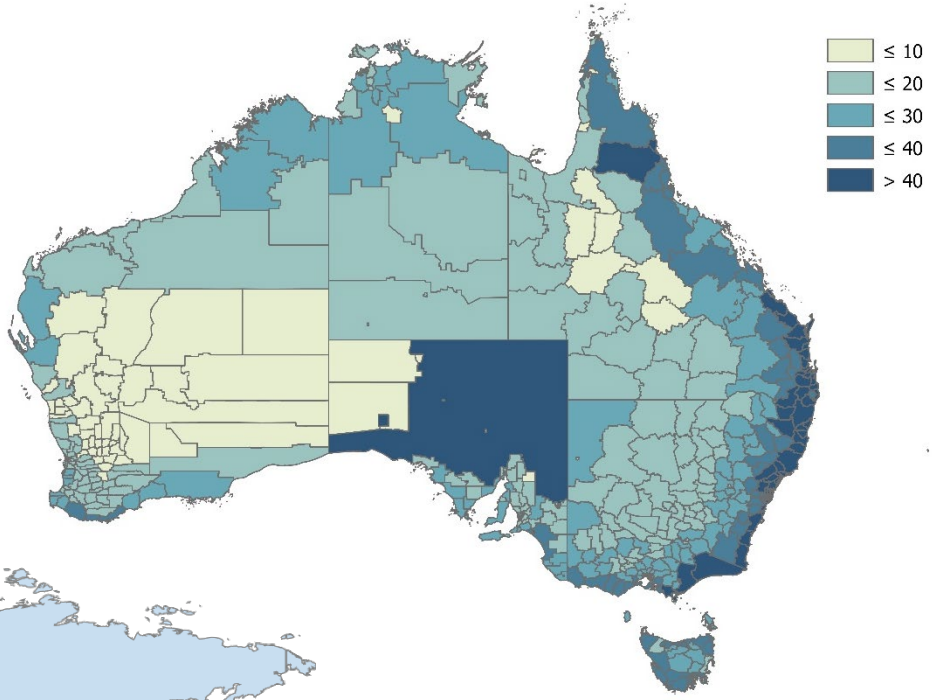
The research question helps inform the appropriate table

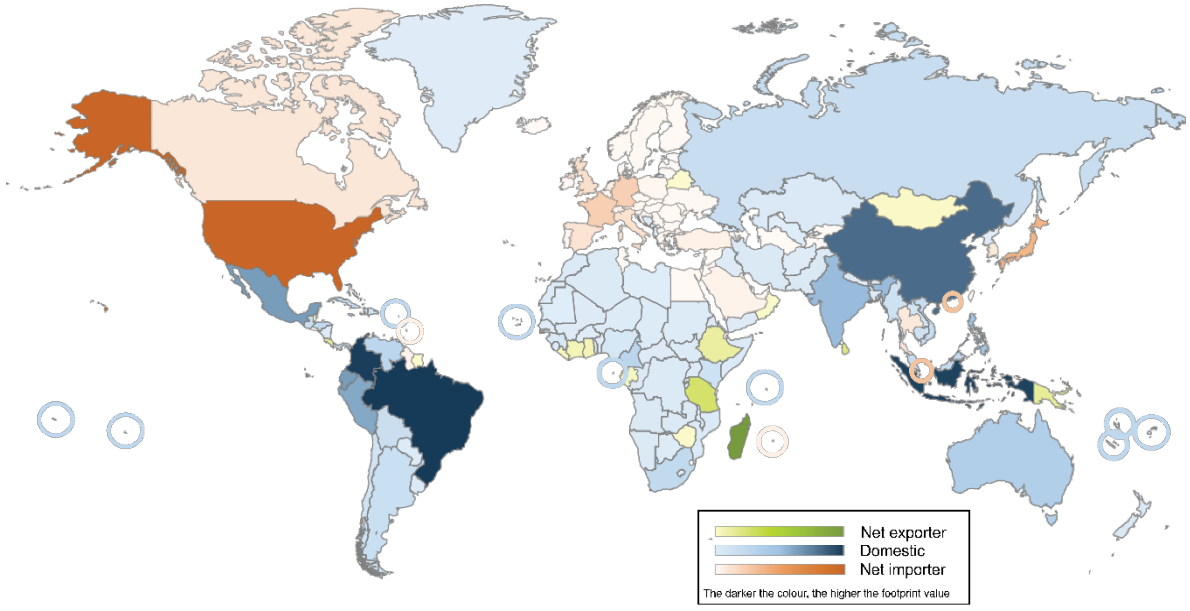
A consumption-based analysis of extinction risk in Australia

AUS IELab

545 local government areas

1284 economic sectors





Quantifying and categorising national extinction-risk footprints

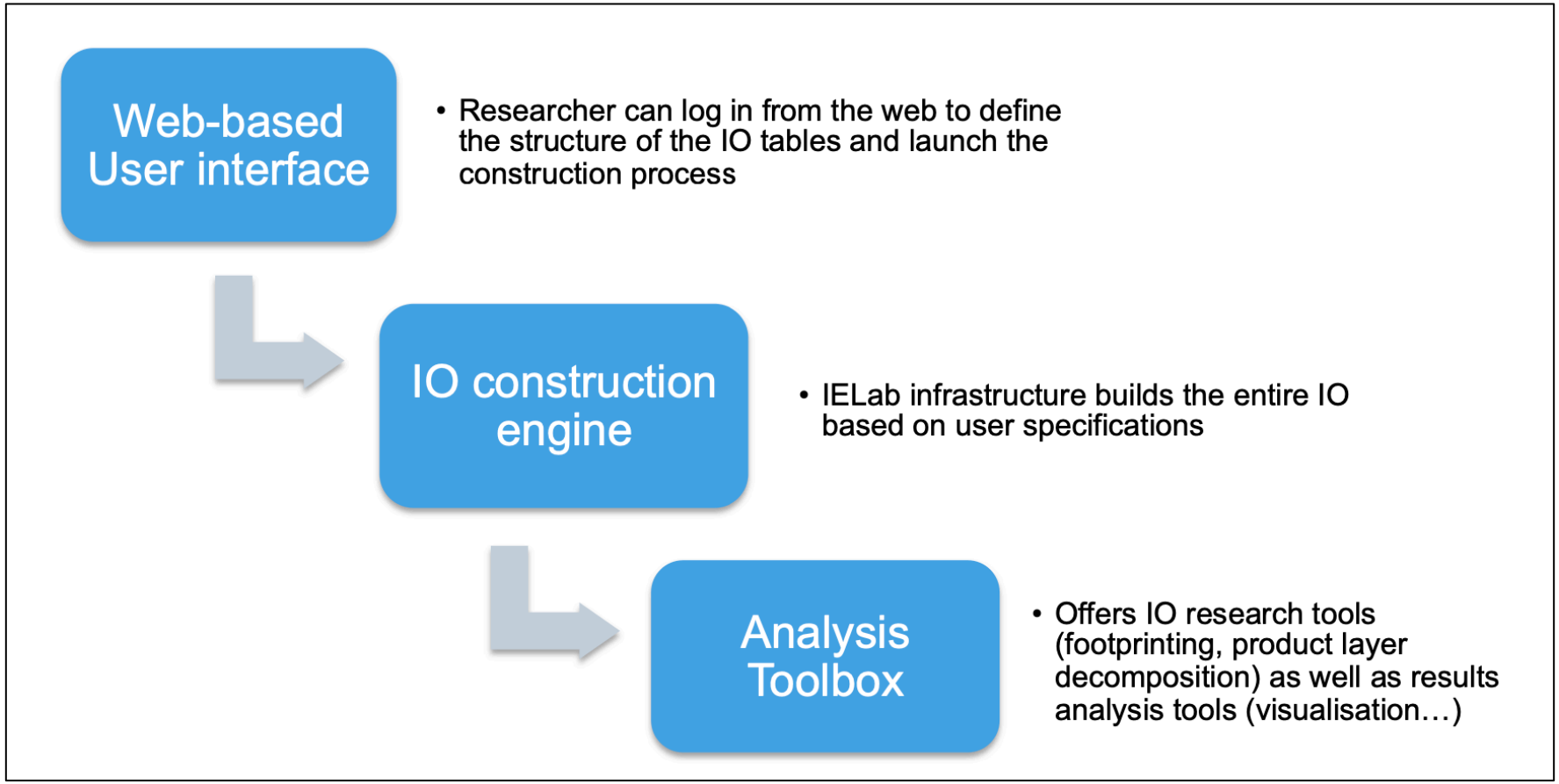
Global IELab

189 countries

26 – 1022 economic sectors per country

How does IE Lab work?

https://www.ielab.info/



IELAB GLOBAL	IELAB AUSTRALIA	IELAB CHINA	IELAB INDONESIA
IELAB JAPAN	IELAB SWEDEN	IELAB TAIWAN	IELAB USA
GLORIA			

164 regions / 97 sectors / 1990-2019 / k USD







Lenzen, M., et al. (2022) Implementing the material footprint to measure progress towards Sustainable Development Goals 8 and 12. Nature Sustainability, 5, 157-166.

IELAB GLOBAL | IELAB AUSTRALIA | IELAB CHINA | IELAB INDONESIA

IELAB JAPAN | IELAB SWEDEN | IELAB TAIWAN | IELAB USA

GLORIA

IndoLab launch - August 2019, Jakarta

- Dr, Futu Faturay – Fiscal Policy Agency
- Dr. Sri Mulyani Indrawati - Minister of Finance of Indonesia



495 regions



1,148 sectors



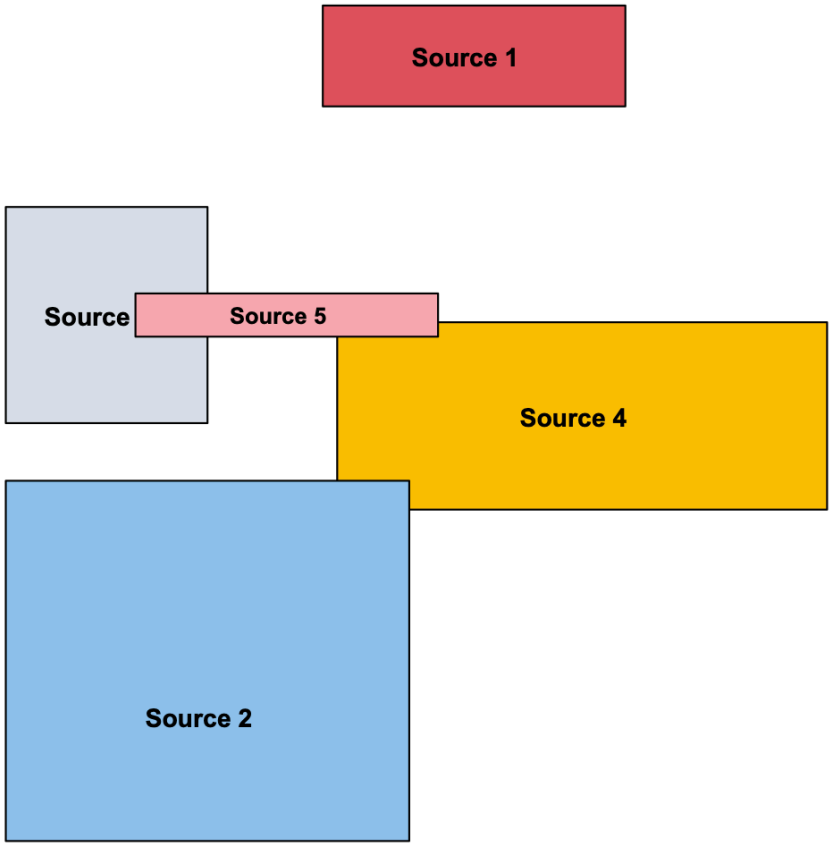
1990-2018





https://www.youtube.com/watch?v=xvpAMm8Tj5U

How does IE Lab work?



Root Classification:

All source data must be "converted" into a highly detailed root classification. All IO tables produced within the IELab are aggregations of this root classification

IE Lab structure

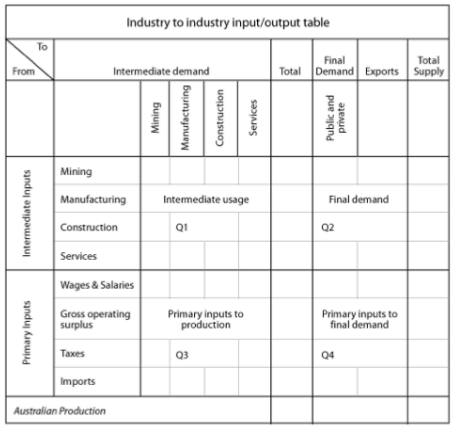
Source

(in native class.)

Source

(in root classifcation)

Final MRIO (base table)

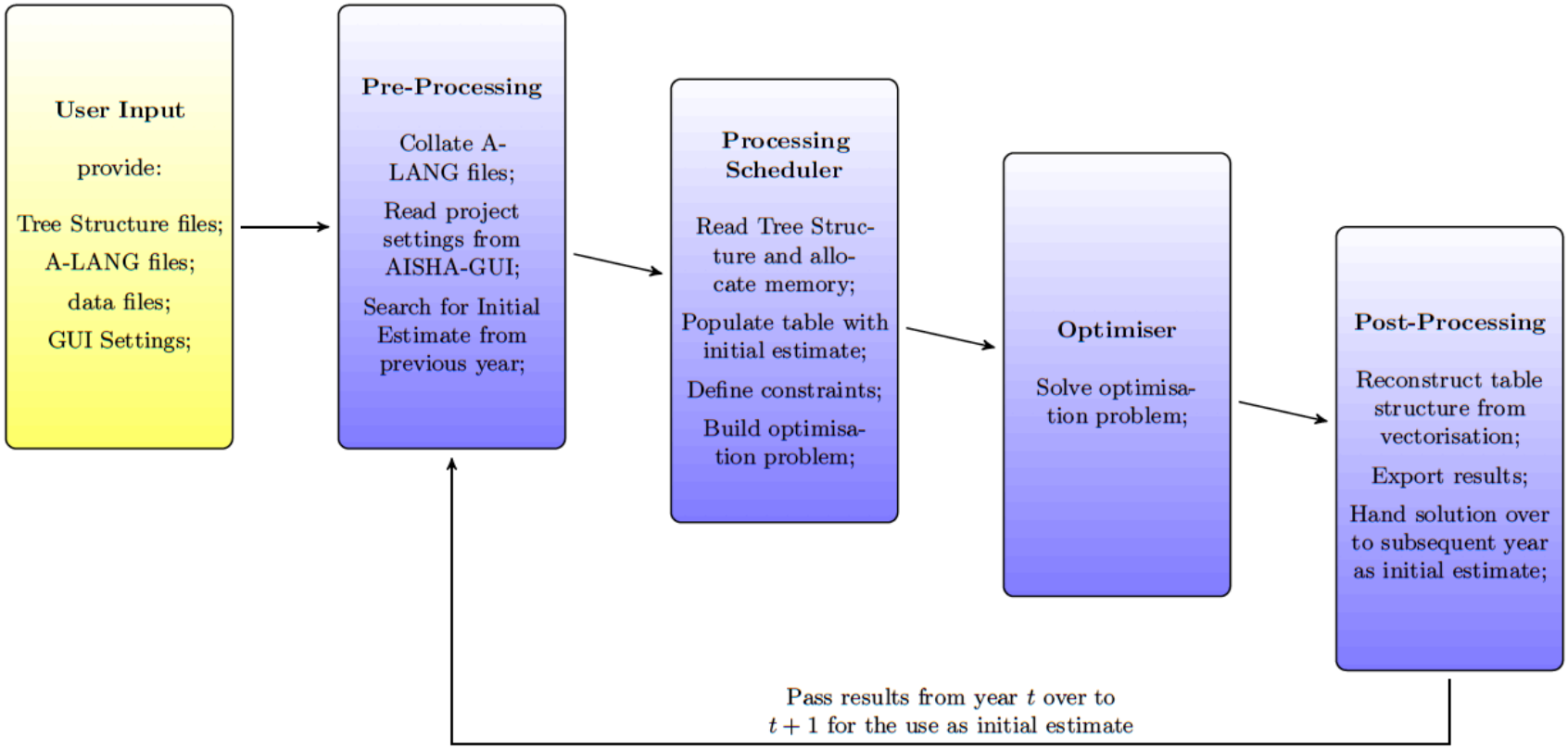


(Built using the source data)

Behind the scenes: IELab-internal data processing

User-controlled IO construction process

IE Lab reconciliation workflow (AISHA)



IE Lab reconciliation workflow (AISHA)

User Input

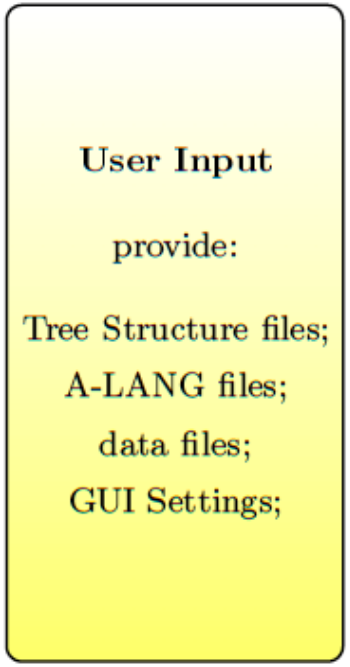
provide:

Tree Structure files;
A-LANG files;
data files;
GUI Settings;

← Tree Structure files:

1. Time (year or financial year), typically between 1 and 50 years;
2. Valuation (basic prices, producers' prices, purchasers' prices, margins, taxes, subsidies, free-on-board, cost-insurance-freight, etc), typically between 3 and 20 valuation types;
3. Region of flow origin (country, state, region, etc), typically between 1 and 250 regions;
4. Entity of flow origin (industry, commodity, or satellite), typically 1, 2 or 3 entities;
5. Sector of flow origin (industries, commodities, or satellite indicators), typically between 20 and 2,000 sectors;
6. Region of flow destination (country, state, region, etc), typically between 1 and 250 regions;
7. Entity of flow destination (industry or commodity), typically 1, 2 or 3 entities;
8. Sector of flow destination (industries or commodities), typically between 20 and 500 sectors.

IE Lab reconciliation workflow (AISHA)



Data files:

For example:

1. National input-output tables, household expenditure survey from ABS
2. GHG emission data from NGGI and ANZSIC database
3. Water use from ABS
4. Energy use from BREE

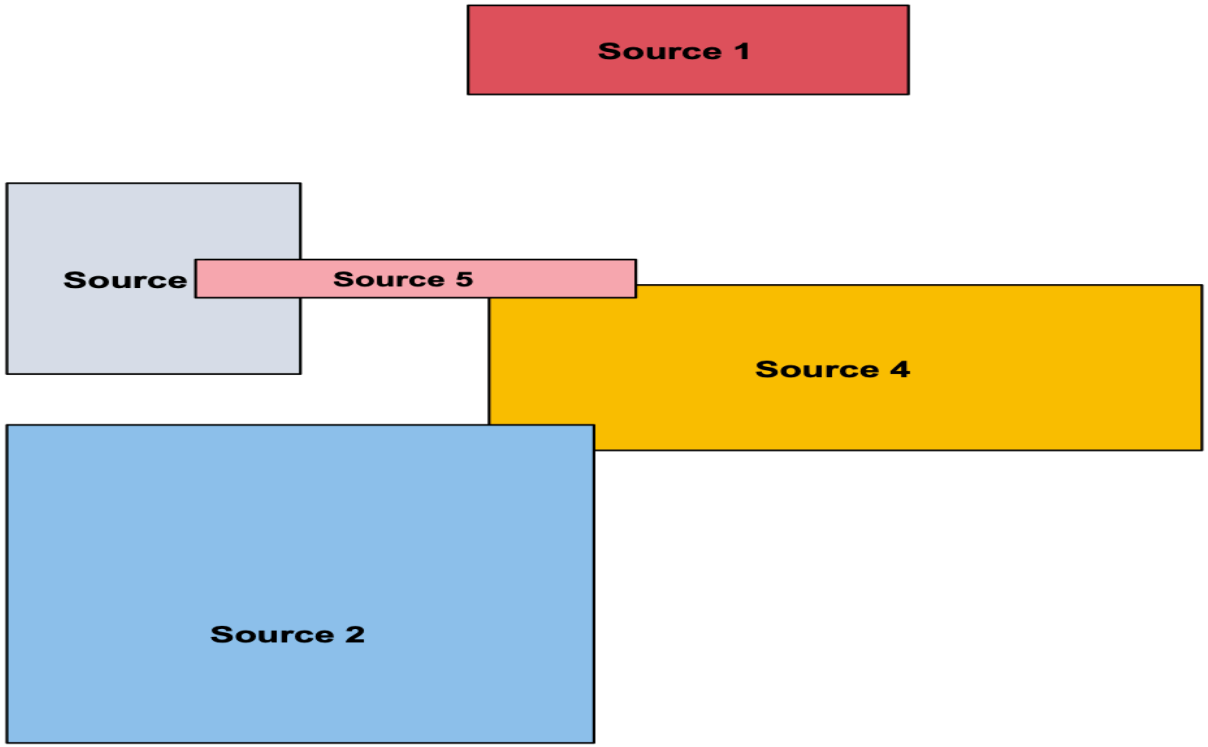
IE Lab reconciliation workflow (AISHA)

User Input

provide:

Tree Structure files;
A-LANG files;
data files;
GUI Settings;

Data files:



IE Lab reconciliation workflow (AISHA)

User Input

provide:

Tree Structure files;
A-LANG files;
data files;
GUI Settings;

A-LANG files:

1. a simple command language
2. formulate all input data for the optimisation problem

Columns 1-10:

1	Incl	#	Parts	Value	Pre-map	Post-map	S.E.	Pre-map	Post-map
IE	Y	1	1	I			E MX1;MN10;CN1e+3;		

Columns 11-19:

Coef 1	Years	Margins	from country	entity	of sector	to country	entity	of sector
A <data filename>	1	1	1	1	1:e	1	2	1:e

IE Lab reconciliation workflow (AISHA)

User Input

provide:

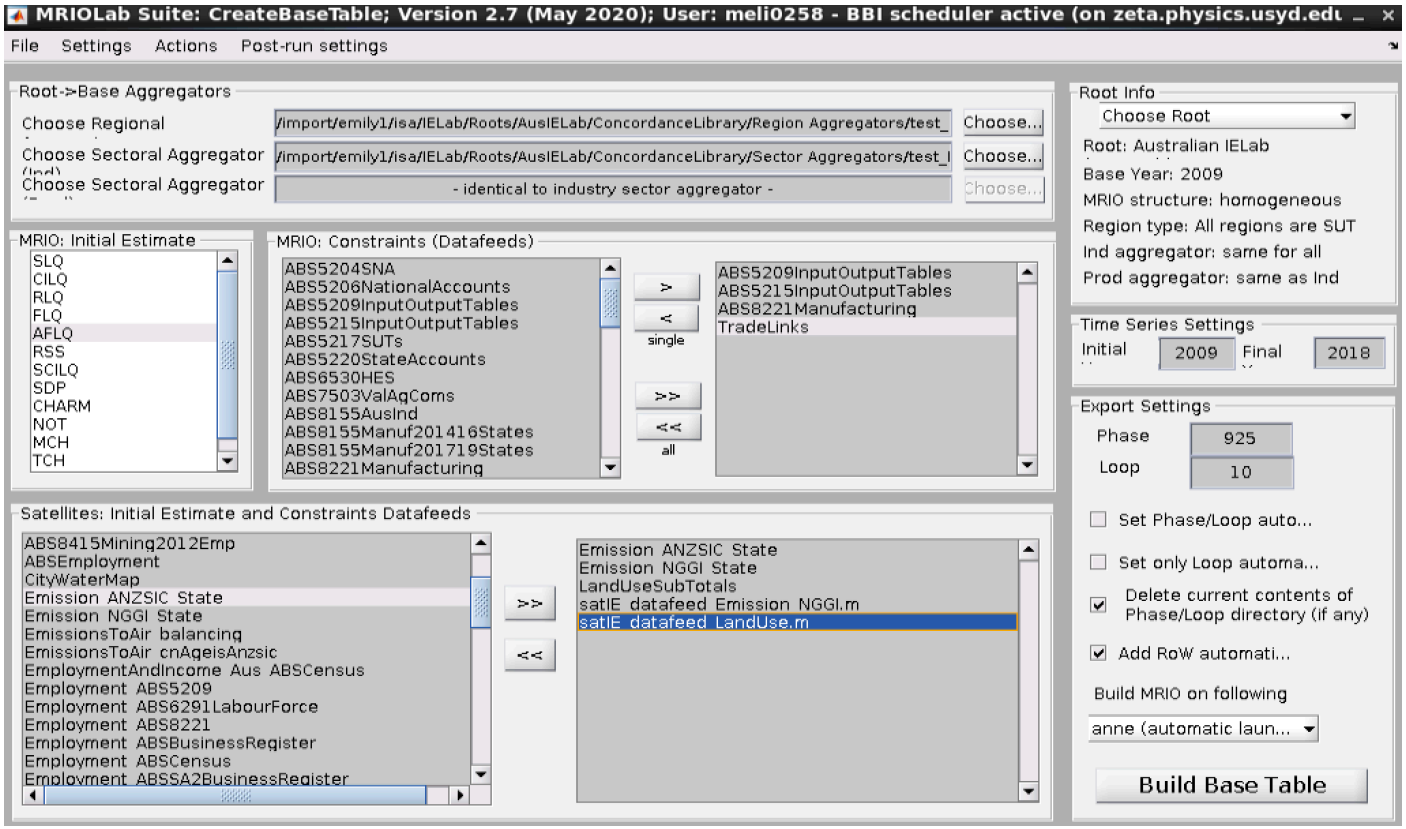
Tree Structure files;

A-LANG files;

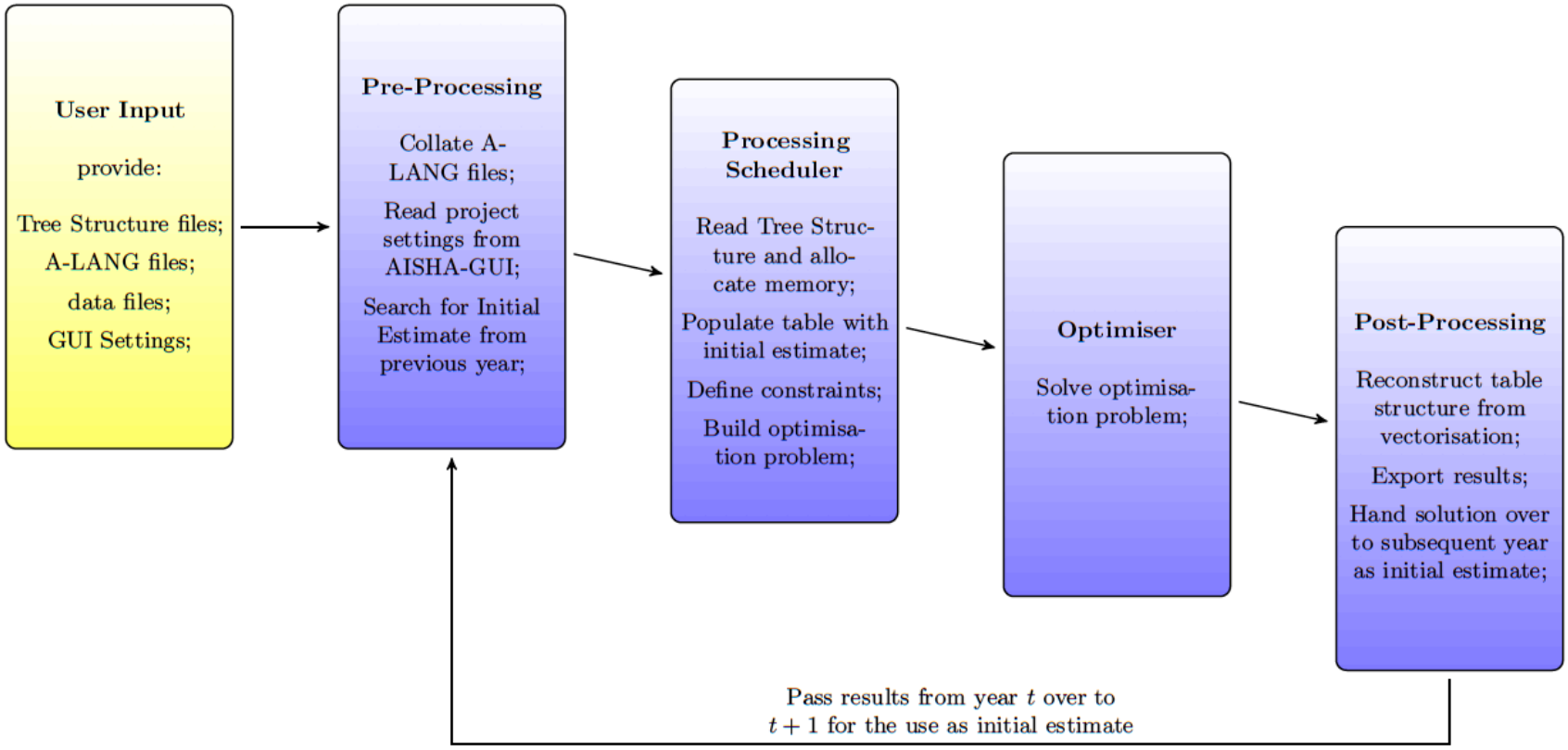
data files;

GUI Settings;

← GUI (graphical user interface) settings

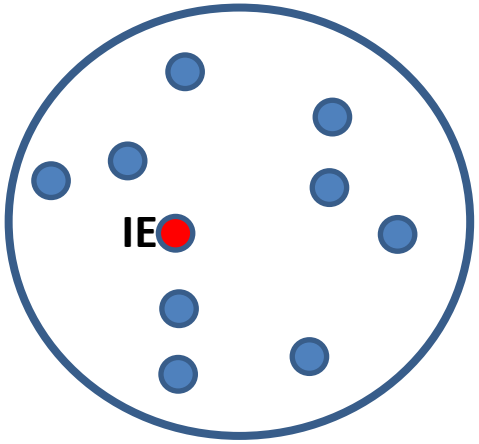


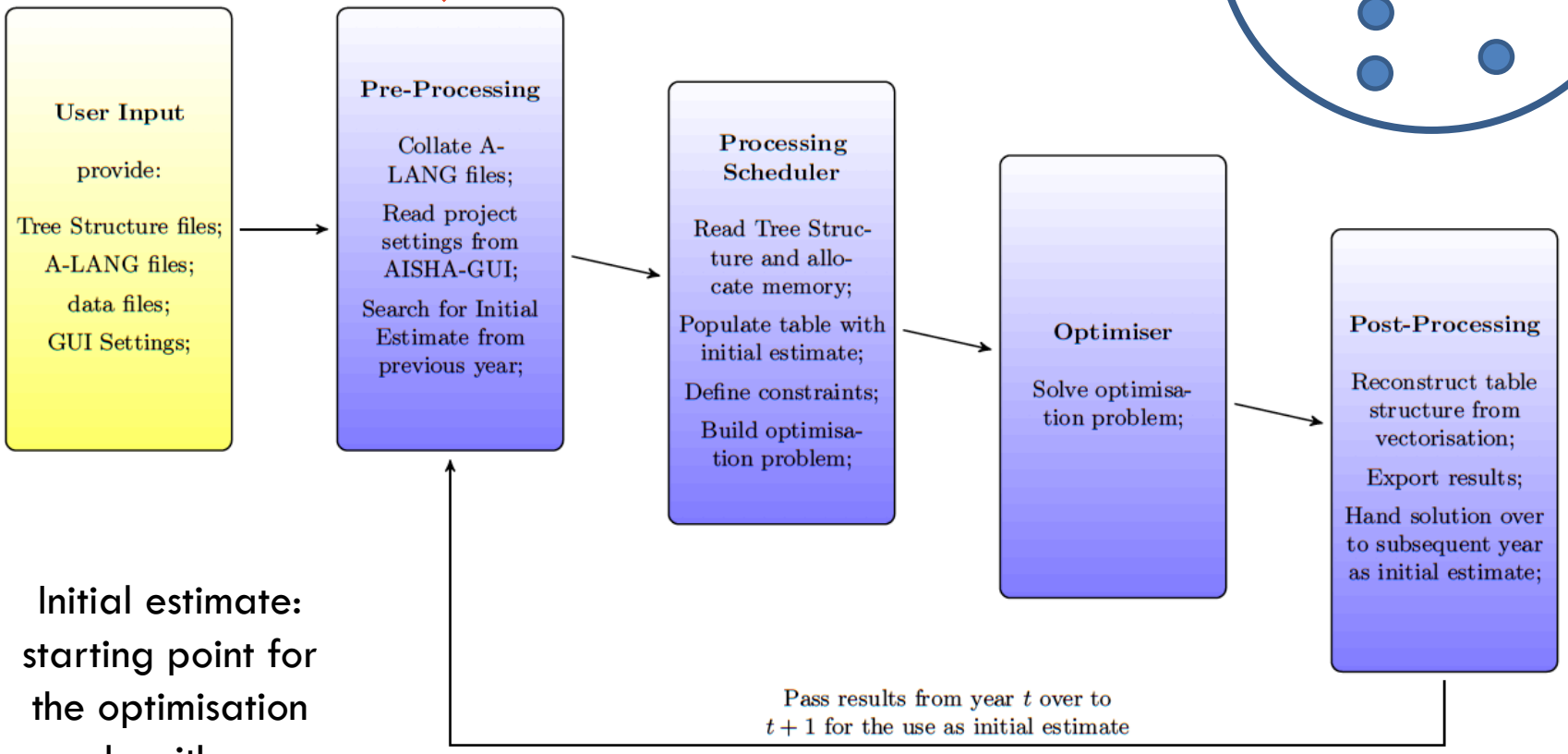
IE Lab reconciliation workflow (AISHA)



IE Lab reconciliation workflow (AISHA)

Pre-processing: prepare the contents for processing scheduler

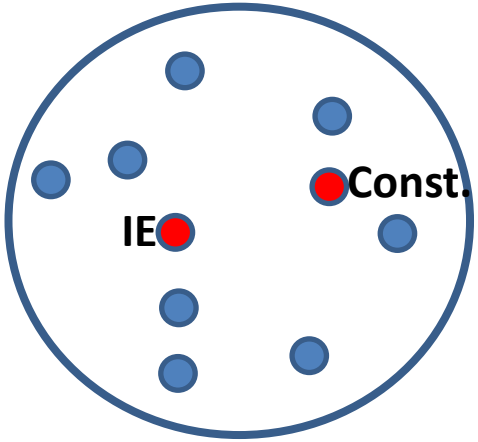


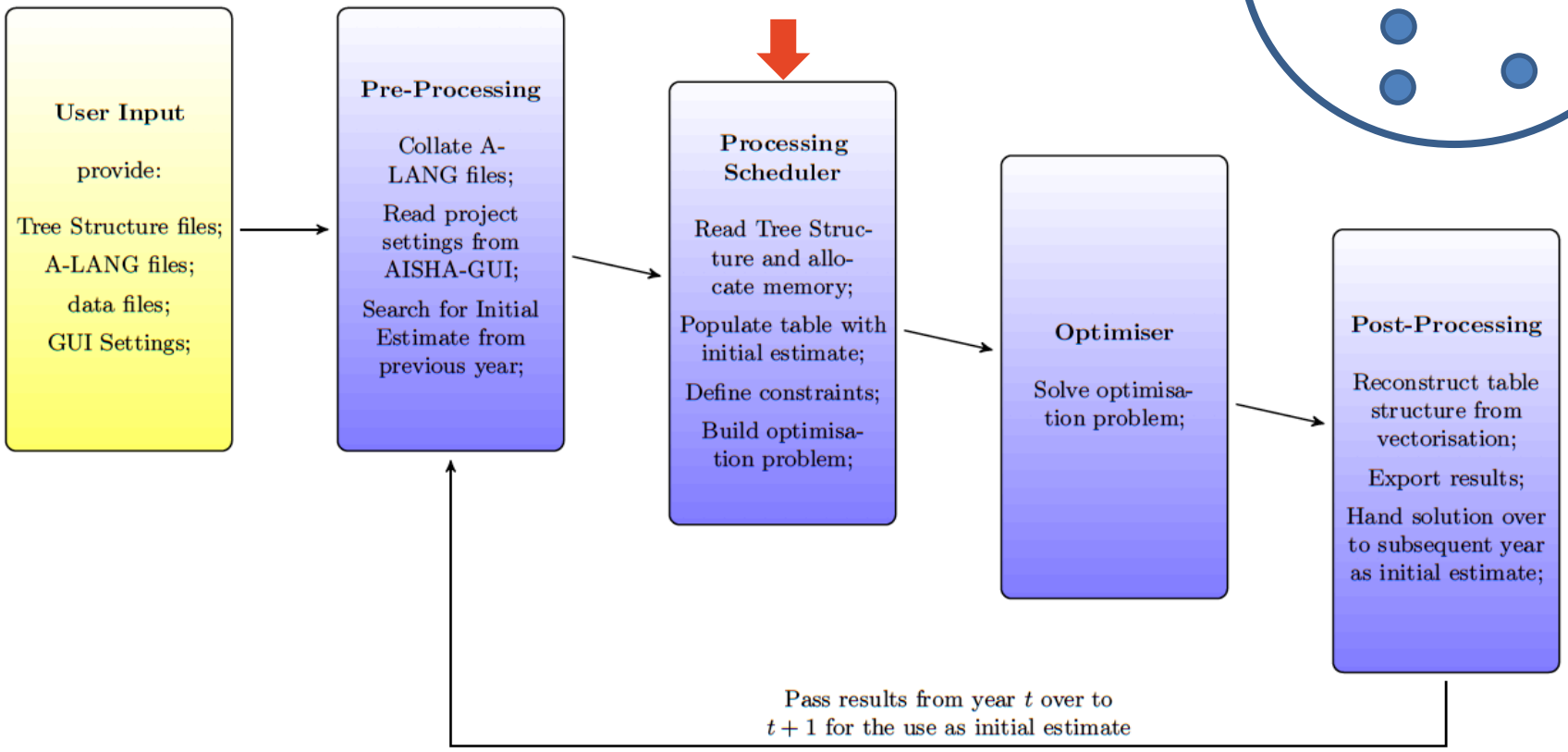


Initial estimate: starting point for the optimisation algorithms

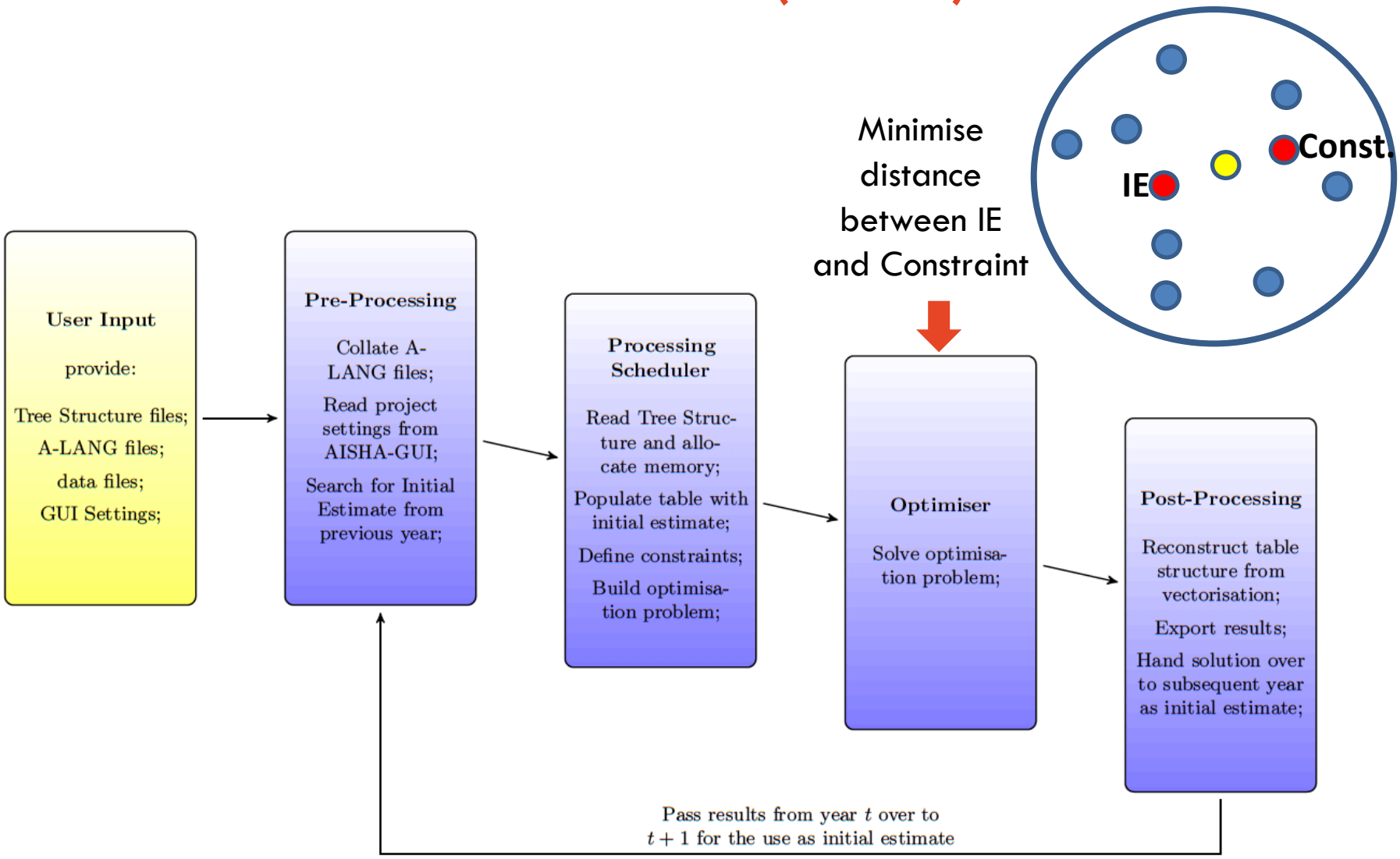
IE Lab reconciliation workflow (AISHA)

Formulate the optimisation problem
Populate IE table
Construct constraints
Read standard error estimate

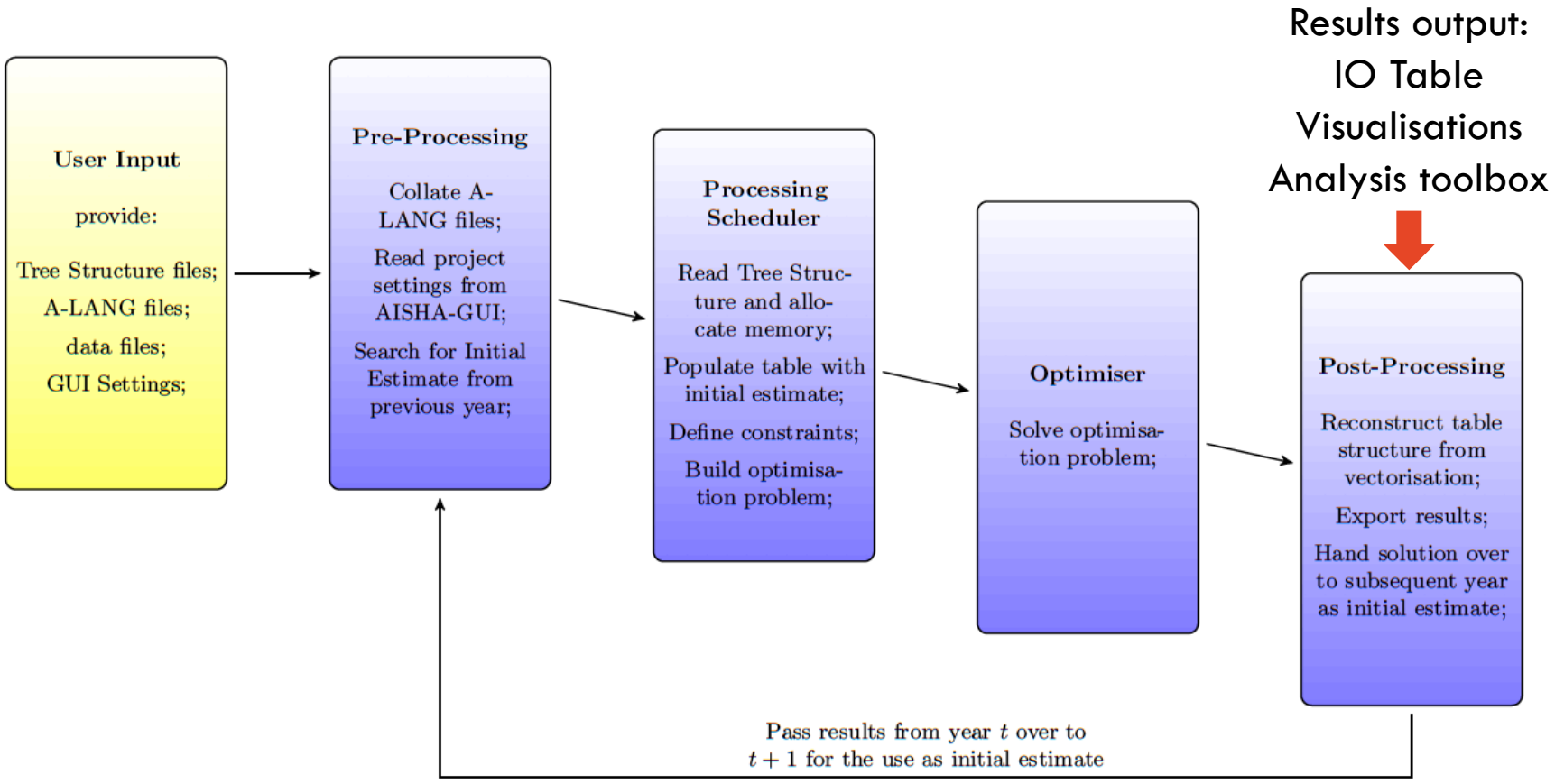




IE Lab reconciliation workflow (AISHA)

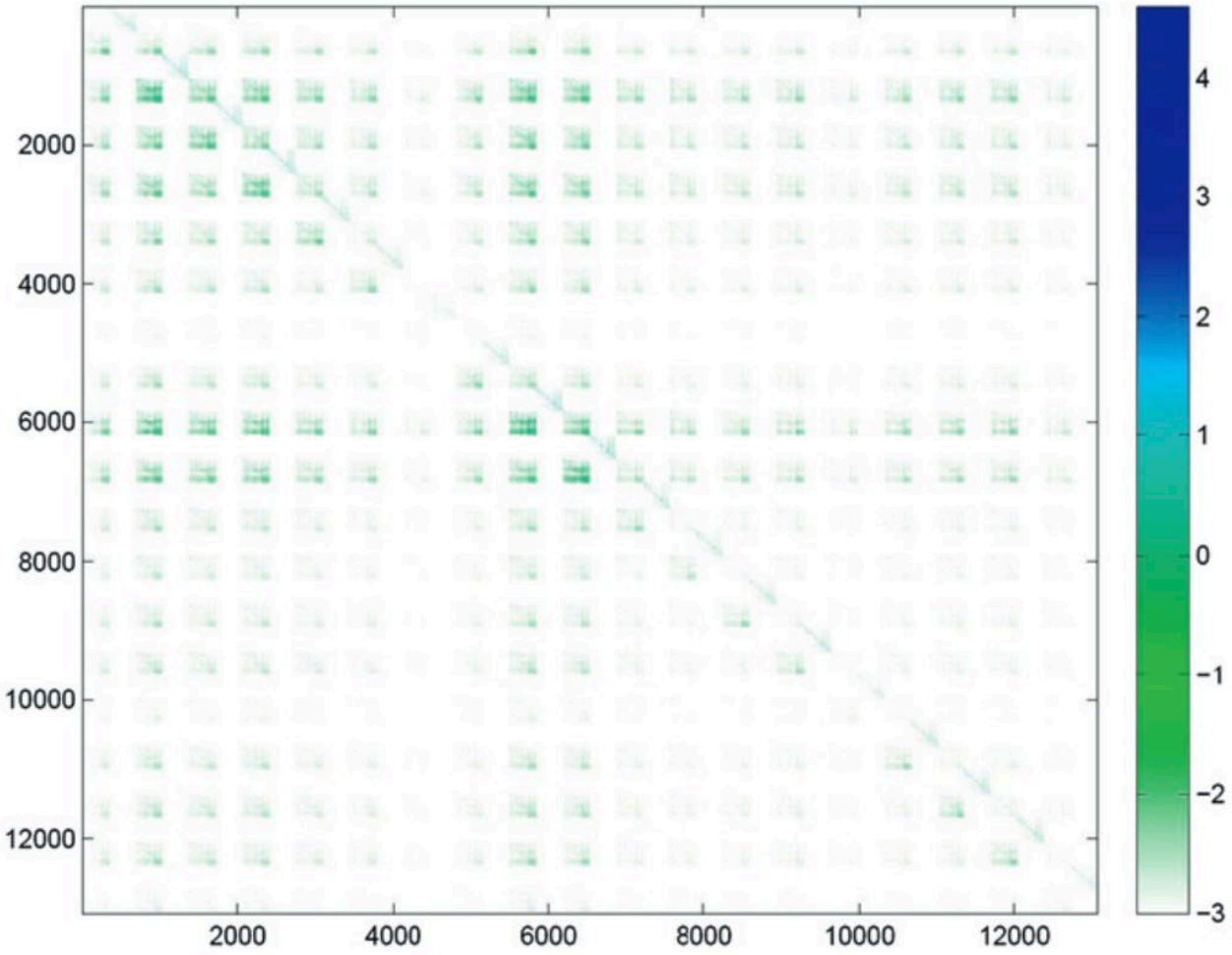


IE Lab reconciliation workflow (AISHA)



IE Lab example output - visualisation

Australian Capital Territory
Victoria
South Australia
Western Australia
Tasmania
Northern Territory
Other Territories
Illawarra
Rest of New South Wales
Brisbane - Morton
Wide Bay - Burnett
Darling Downs
South-West
Fitzroy
Central West
Mackay
Northern
Far North
North-West



IE Lab example output – uncertainty analysis



$$F = m \times a$$

Where *m* denotes mass (kg) and *a* denotes acceleration (ms^{-2})

100 times

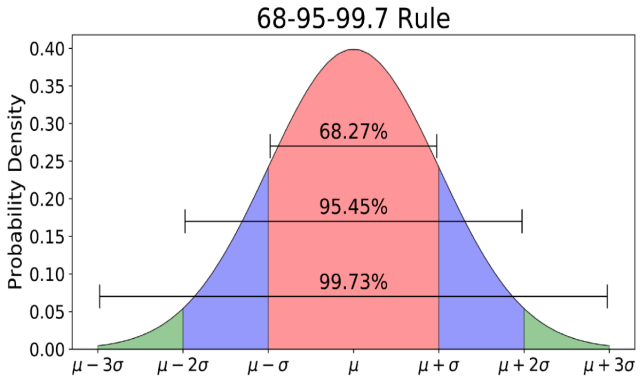
Uncertainties in *m* and *a*

Uncertainties in *F*

Mean & Variance

Confidence intervals

Monte-carlo simulation



68% of the data is within 1 standard deviation, 95% is within 2 standard deviation, 99.7% is within 3 standard deviations

Uncertainties in MRIO analysis

UNCERTAINTY ANALYSIS FOR MULTI-REGION INPUT–OUTPUT MODELS – A CASE STUDY OF THE UK'S CARBON FOOTPRINT

Manfred Lenzen , Richard Wood & Thomas Wiedmann

To cite this article: Manfred Lenzen , Richard Wood & Thomas Wiedmann (2010) UNCERTAINTY ANALYSIS FOR MULTI-REGION INPUT–OUTPUT MODELS – A CASE STUDY OF THE UK'S CARBON FOOTPRINT, Economic Systems Research, 22:1, 43-63, DOI: 10.1080/09535311003661226

To link to this article: https://doi.org/10.1080/09535311003661226

Published online: 13 May 2010.

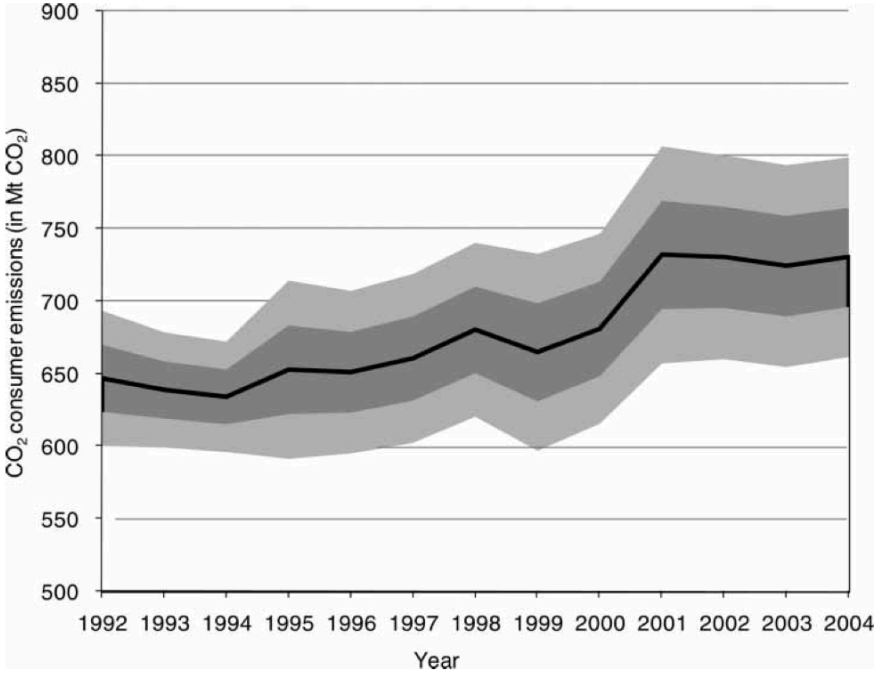
Uncertainties in MRIO analysis

Footprint = **qLY**

- Uncertainties of Raw Data (**Q, T, V, Y**)
- Uncertainties of CO_2 data

Monte-Carlo simulation

FIGURE 9. UK consumer CO_2 emissions between 1992 and 2004.



Note: Uncertainty ranges are presented as confidence intervals of width σ (+/−1SD, 67% of all observations) and 2σ (+/−2SD, 95% of all observations).

Tutorial



THE UNIVERSITY OF
SYDNEY

Tutorial Activity

Structure of multi-region input-output (MRIO) tables

Construct a multi-region input-output (MRIO) table featuring two regions. Your second region should have a different number of sectors from the first region. Make sure you include the ROW import and ROW export vectors in appropriate place in the table.

The key is to get the structure correct.

How do I add values? Two options:

- Add dummy values to complete and balance the table; or
- Get data from the ATLAS of economic complexity (and then balance the table)

(https://atlas.cid.harvard.edu/explore)

Homework for Week 7

Homework week 7 (Due: October 28, 2026)

Read all questions before you start on the Homework

Use your 14-sector table to complete the following:

1. **Add a second region to your input-output system**
2. **Use a different number of sectors for your second region than for your first region**

This means that the second region should not have 14 sectors in the T matrix. You need to choose a number other than 14.

3. **Imports and Exports**

Make sure that you remove your original export vector from your existing final demand matrix and your original import vector from your existing value-added matrix, and only use one Rest of World (ROW) import and one Rest of World (ROW) export vector as discussed in the lecture.

4. **Balance your MRIO.**

Same idea: total input = total output for each sector

** **Extension task (not assessed)**

Extend your Satellite matrix to your second region, calculate q, then calculate

1. The footprints of the commodities bought by Region B from Region A (i.e. commodities are produced in Region A, but consumed in Region B).
2. The footprints of the commodities bought by Region A from Region B (i.e. commodities are produced in Region B, but consumed in Region A).
3. Find another footprint that interests you

After this week's lecture, we will go back to the single region IO table (SRIO) that you have used so far.

EXPLANATION: Explain why you only have one import row in the value-added matrix (V) and one export column in the final demand matrix (Y), even though there is more than one region included in the IO table (100 words). Imagine you need to calculate the footprint of Australia's final consumption of washing machines, where the supply chains originate in China, Germany and South Korea. What data would you need for doing such an assessment? (100 words)

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- The structure of a MRIO is different from that of a SRIO
- Treatment of imports and exports
- Multi-region input-output tables enable more sophisticated footprint calculations
- How MRIO tables are constructed

More to come:

- Guest lecture on footprint applications